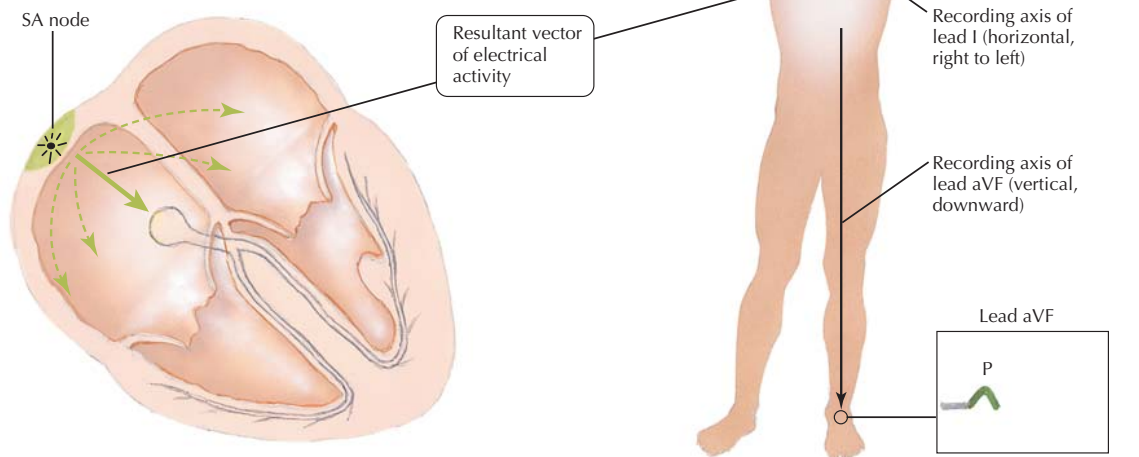


Normal Sequence of Cardiac Depolarization and Repolarization and Derivation of ECG

A. Impulse origin and atrial depolarization

Impulse originates at SA node, and wave of depolarization spreads over atria, resulting in electrical vector directed downward and to left. This causes upward (positive) deflection in ECG tracing in leads I and aVF (P wave).



B. Septal depolarization

After brief delay at AV node, impulse traverses common bundle of His and right and left bundle branches and then enters interventricular septum, causing myocardial depolarization with electrical vector directed to right and downward. This results in small negative (downward) deflection in lead I (Q wave) and positive (upward) deflection in lead aVF (R wave).

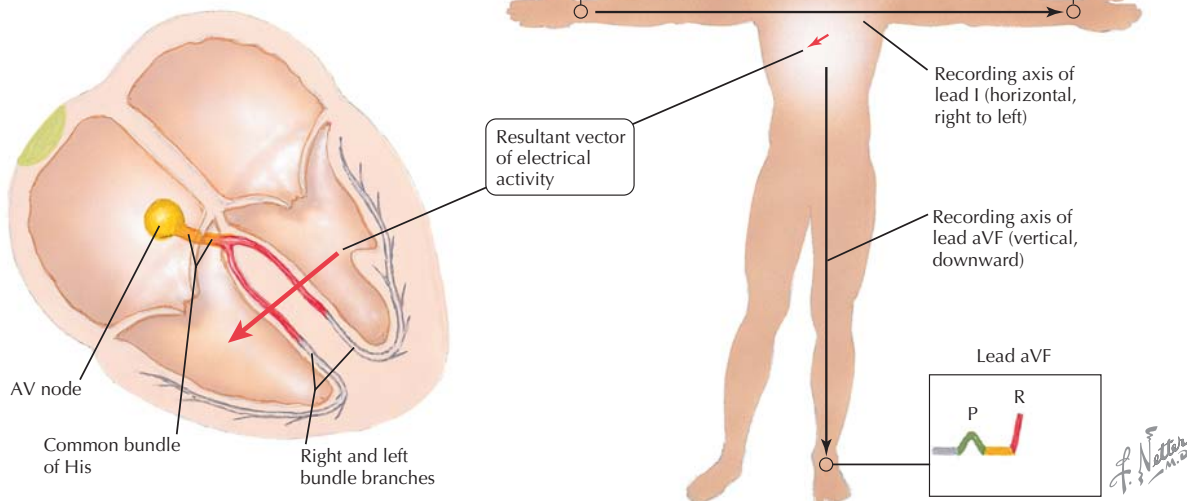
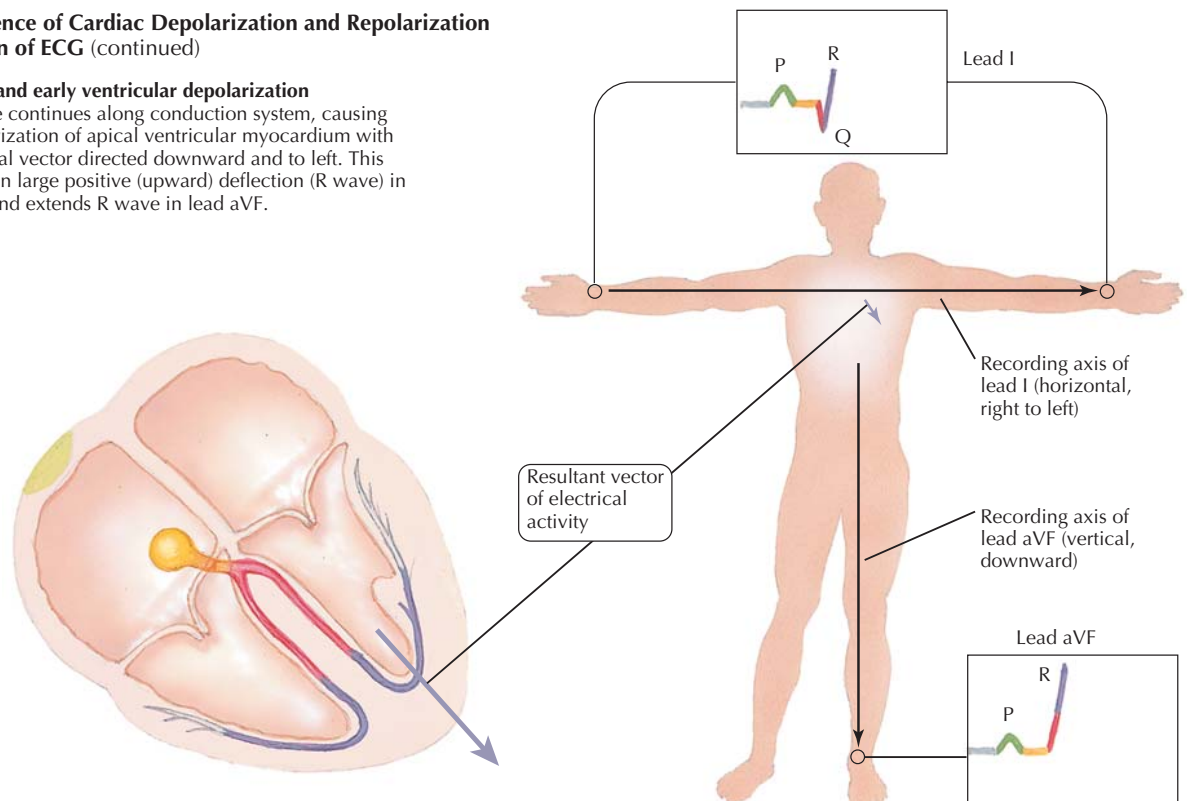


Figure 9.2 Cardiac Depolarization and Repolarization Part I This series of figures (Figs. 9.2 to 9.4) depicts the sequence of cardiac electrical events during the cardiac cycle and the surface recordings that result from these events and constitute the ECG. The initial events are depicted above, in which the firing of the SA node results in depolarization of the atria, recorded on the ECG as the P wave. This depolarization spreads to the AV node, the bundle of His, and bundle branches, with the resultant electrical activity vectors and surface ECG recordings. Note that the morphology of the ECG recording varies with lead placement.

Normal Sequence of Cardiac Depolarization and Repolarization and Derivation of ECG (continued)

C. Apical and early ventricular depolarization

Impulse continues along conduction system, causing depolarization of apical ventricular myocardium with electrical vector directed downward and to left. This results in large positive (upward) deflection (R wave) in lead I and extends R wave in lead aVF.



D. Late ventricular depolarization

As depolarization progresses over ventricles, vector shifts to become directed superiorly as well as to left, thus extending upward R wave in lead I and causing negative (downward) deflection (S wave) in lead aVF.

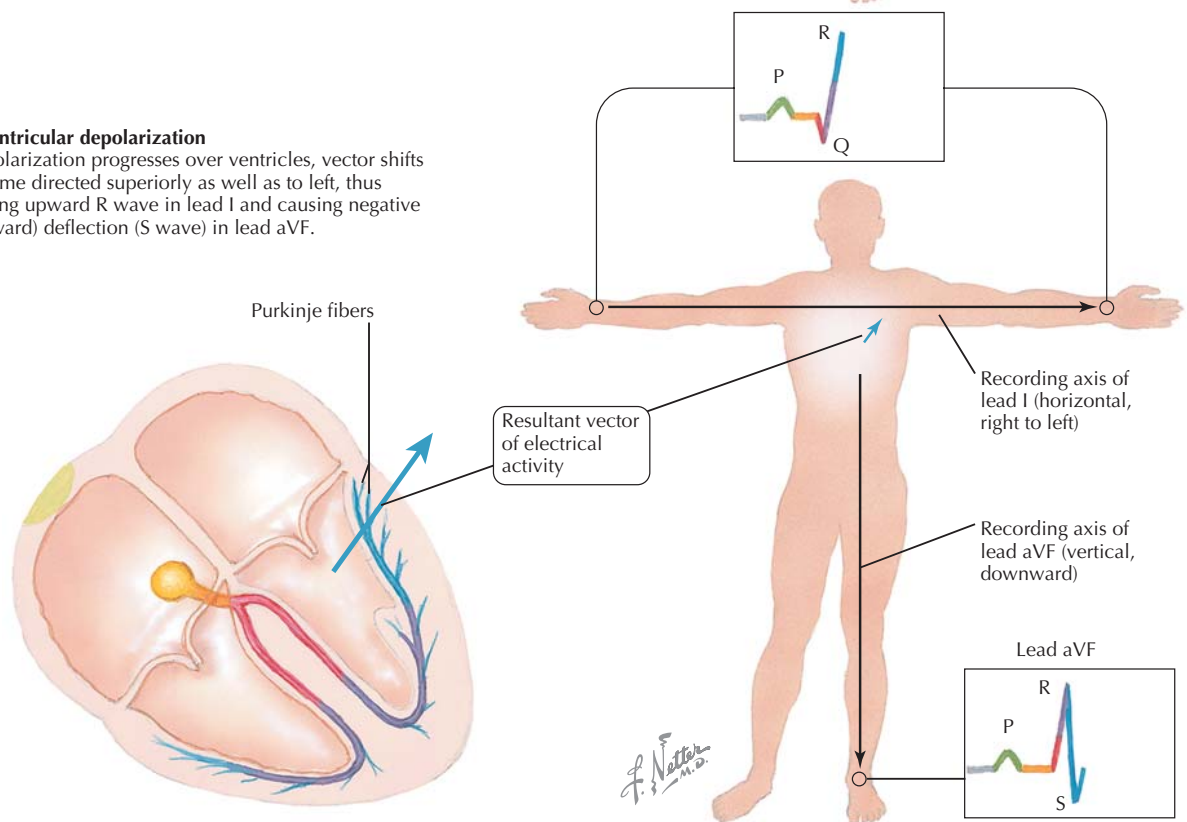
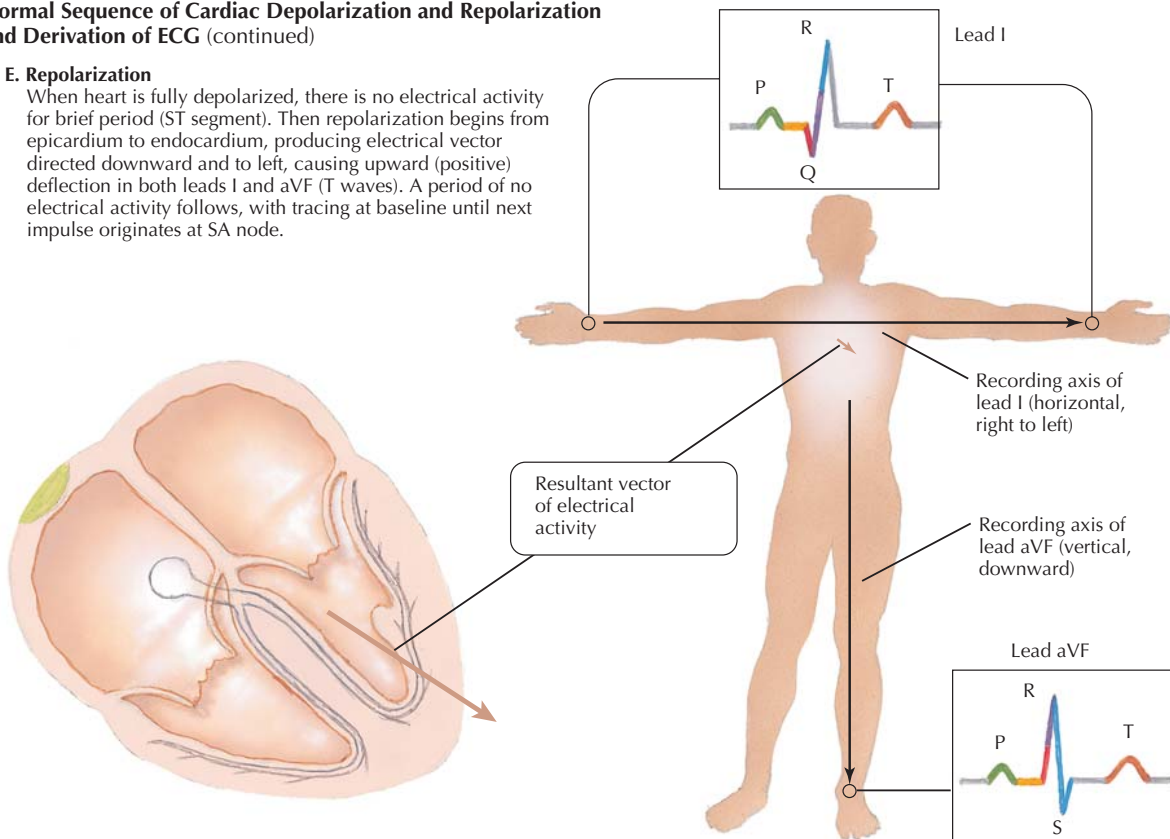


Figure 9.3 Cardiac Depolarization and Repolarization Part II Septal depolarization (see Fig. 9.2) is followed by apical and ventricular wall depolarization; resultant vectors of electrical activity produce the QRS complex of the ECG.

Normal Sequence of Cardiac Depolarization and Repolarization and Derivation of ECG (continued)

E. Repolarization

When heart is fully depolarized, there is no electrical activity for brief period (ST segment). Then repolarization begins from epicardium to endocardium, producing electrical vector directed downward and to left, causing upward (positive) deflection in both leads I and aVF (T waves). A period of no electrical activity follows, with tracing at baseline until next impulse originates at SA node.



F. Summary of cardiac electrical activity

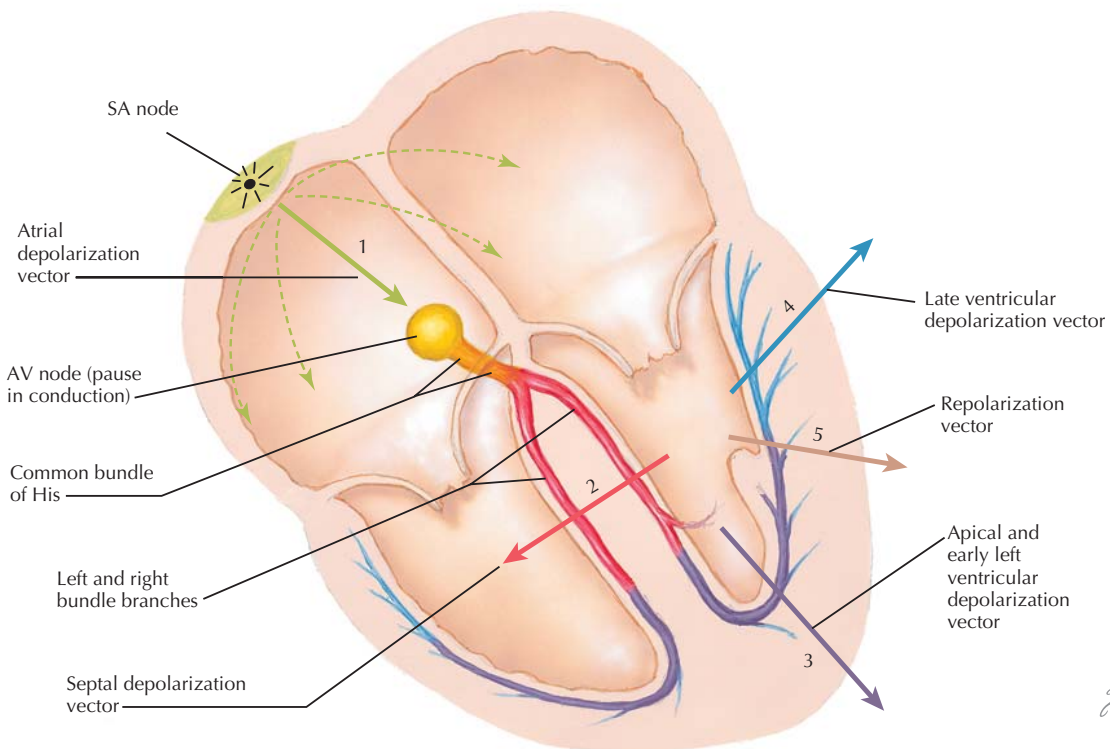


Figure 9.4 Cardiac Depolarization and Repolarization Part III Ventricular depolarization (see Fig. 9.3) is followed by ventricular repolarization; the resultant vector of electrical activity produces the T wave of the ECG. The cardiac electrical activity and resultant vectors (1-5) through the cycle are summarized in F.

CHAPTER 10

Flow, Pressure, and Resistance

Hemodynamics is the study of the forces involved in circulation of blood. Although arterial blood pressure is a convenient and readily measured parameter, understanding the broader hemodynamic state of an individual is essential when assessing cardiovascular disease.

BASIC HEMODYNAMICS

The rate of flow through the circulation (Q) is determined by the pressure gradient across the circulation (ΔP) and the resistance of the circulation (R):

$$Q = \Delta P / R$$

Pressure is defined as force per unit area. For example, in an auto tire, pressure is measured as pounds per square inch (PSI). The unit for measurement of blood pressure is most often mm Hg (1 mm Hg = 1 torr), and the pressure gradient (ΔP) for flow is arterial pressure minus venous pressure. Resistance is the impedance to flow and can be measured in units of mm Hg/mL/min. Resistance can be quantified as the pressure rise associated with an incremental rise in flow. The greatest resistance in the circulation occurs in the smallest arteries and arterioles (see Fig. 8.1). Factors that determine resistance are considered later in the chapter (see “Biophysics of Circulation”).

BLOOD PRESSURE

The blood pressures in the venous systems of both pulmonary and systemic circulations are considerably lower than pressures in the respective arterial systems. In addition, blood pressures in the pulmonary circulation are lower than corresponding pressures in the systemic circulations. Systemic arterial pressure at rest is normally about 120/80 mm Hg (systolic/diastolic), compared with 25/10 mm Hg for pulmonary artery pressure. Some important definitions related to arterial pressure are as follows:

- **Systolic arterial pressure:** Peak arterial pressure reached during ejection of blood by the heart.
- **Diastolic arterial pressure:** Lowest arterial pressure reached during diastole, while the heart is relaxed and filling (not ejecting blood).



Flow, pressure, and resistance can be related in the formula

$$Q = \Delta P / R$$

where Q is flow, ΔP is the pressure gradient, and R is resistance to flow.

Rearranging this formula,

$$\Delta P = QR$$

This equation is analogous to Ohm's law, which states that $V = IR$, where V is electrical potential (electrical gradient), I is current (electrical flow), and R is impedance (electrical resistance).

- **Arterial pulse pressure:** The difference between systolic and diastolic pressures; dependent on stroke volume (volume ejected by one ventricle during one contraction) and arterial compliance.
- **Mean arterial pressure (MAP):** The average pressure over a complete cardiac cycle of systole and diastole; dependent on peripheral resistance and cardiac output (volume ejected by one ventricle per unit time).

Mean arterial pressure is not the simple arithmetic mean of systolic and diastolic pressures, because of the irregular shape of the arterial pressure curve (Fig. 10.1). MAP can be approximated by adding one third of the pulse pressure to the diastolic pressure.



Physiological pressures are usually given in units of mm Hg or cm H₂O. In other words, 1 mm Hg is the pressure that would support a column of mercury (Hg) at a height of 1 millimeter (mm), and 1 cm H₂O is the pressure that would support a column of water 1 centimeter (cm) high. Another way to conceptualize these units is that a column of water 1 cm high would exert 1 cm H₂O pressure at its base, and a column of mercury 1 mm high would exert 1 mm Hg pressure at its base. Although these may appear to be unconventional units for quantifying pressure, they are useful because water or mercury manometers are often used to measure pressure. In some texts, the unit *torr* is used; it is equal to 1 mm Hg.

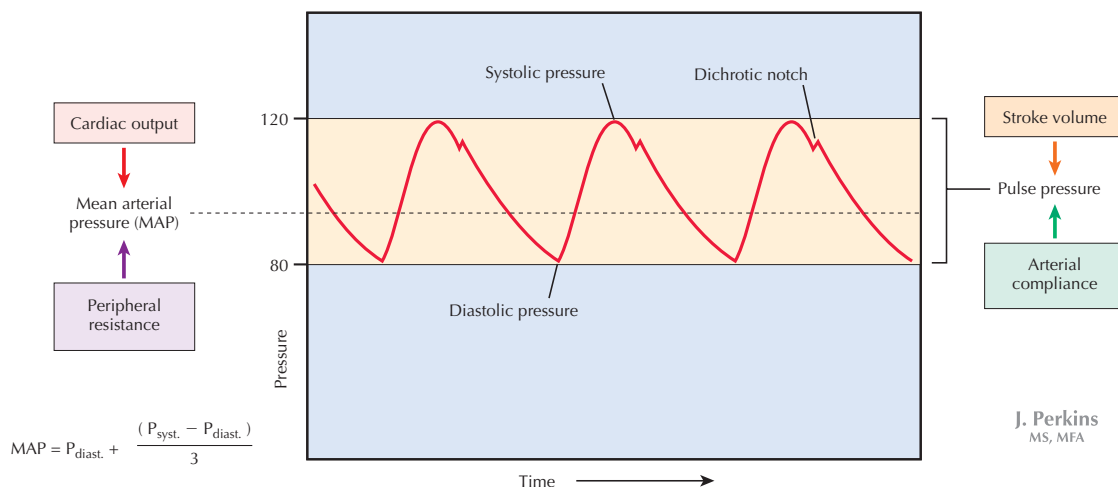


Figure 10.1 The Arterial Pressure Wave The arterial pressure wave represents the changes in pressure in the arterial system over periods of systole, during which the stroke volume is ejected from the left ventricle, and diastole, during which the heart is refilling and blood in the arterial system continues to flow downstream. Arterial pressure is affected by cardiac output, stroke volume, arterial compliance, and peripheral resistance. Mean arterial pressure can be approximated based on the formula shown. The first upward deflection in the curve marks the beginning of systole, the period of cardiac ejection. The transient irregularity in the downward slope of the wave is known as the dicrotic notch, which is produced by closure of the aortic valve, marking the beginning of diastole. During diastole, the heart refills, while blood in the arterial system runs downstream, reducing arterial pressure.

Pressure waves vary through the cardiovascular system (Fig. 10.2). Both high and low pressures are encountered in the ventricles (approximately 120/0 and 25/0 mm Hg at rest for left and right ventricles, respectively). High systolic pressure is necessary for pumping of blood through the circulation, whereas the low pressures are required for return of blood to the heart during diastole. Note that while mean pressures in the large arteries are somewhat lower than in the aorta, pressure pulsations are greater in larger arteries. This phenomenon is mainly attributable to two factors: changes in pressure travel more rapidly downstream than the actual blood flow, which accentuates the pulsatility downstream, and pressure changes are reflected back at branch points, again accentuating downstream pulsatility.

BIOPHYSICS OF CIRCULATION

Blood flow through vessels is a complex phenomenon, involving a nonhomogenous fluid flowing in a pulsatile manner through distensible, branching tubes of various dimensions. Under most conditions, this flow can be described by Poiseuille's law:

$$Q = \frac{\Delta P \pi r^4}{\eta 8L}$$

where Q is flow, ΔP is the pressure gradient from one end of a tube to the other, r^4 is the radius of the tube to the fourth power, η is the viscosity of the fluid, and L is the length of the tube. The effects of this relationship are illustrated in Figure 10.3.

Based on Poiseuille's law, flow (Q) through a tube will be:

- Directly proportional to the longitudinal pressure gradient (inflow pressure minus outflow pressure).
- Inversely proportional to the length of the tube.
- Inversely proportional to the viscosity of the fluid.
- Directly proportional to the fourth power of the radius of the tube.

For example, if the radius of a tube is doubled, flow will be increased by a factor of 16, assuming the pressure gradient is maintained. Physiological regulation of regional blood flow on a moment-to-moment basis mainly involves changes in radius (vasodilation and vasoconstriction) of the small arteries and arterioles, taking advantage of this powerful factor. Under normal circumstances, viscosity of blood is not an issue; however, changes in hematocrit are associated with large changes in blood viscosity, as occur in anemia and polycythemia.

Because $Q = \Delta P/R$, resistance can be described as:

$$R = \frac{\eta 8L}{\pi r^4}$$



Of the factors affecting flow through a tube, the most important is the radius of the tube. While flow is inversely proportional to the length of the tube and the viscosity of the fluid, and directly proportional to the hydrostatic pressure gradient in the tube, it is directly proportional to the fourth power of the radius of the tube. Therefore, doubling the radius of the tube will cause a 16-fold increase in flow, if other factors are constant.

CLINICAL CORRELATE

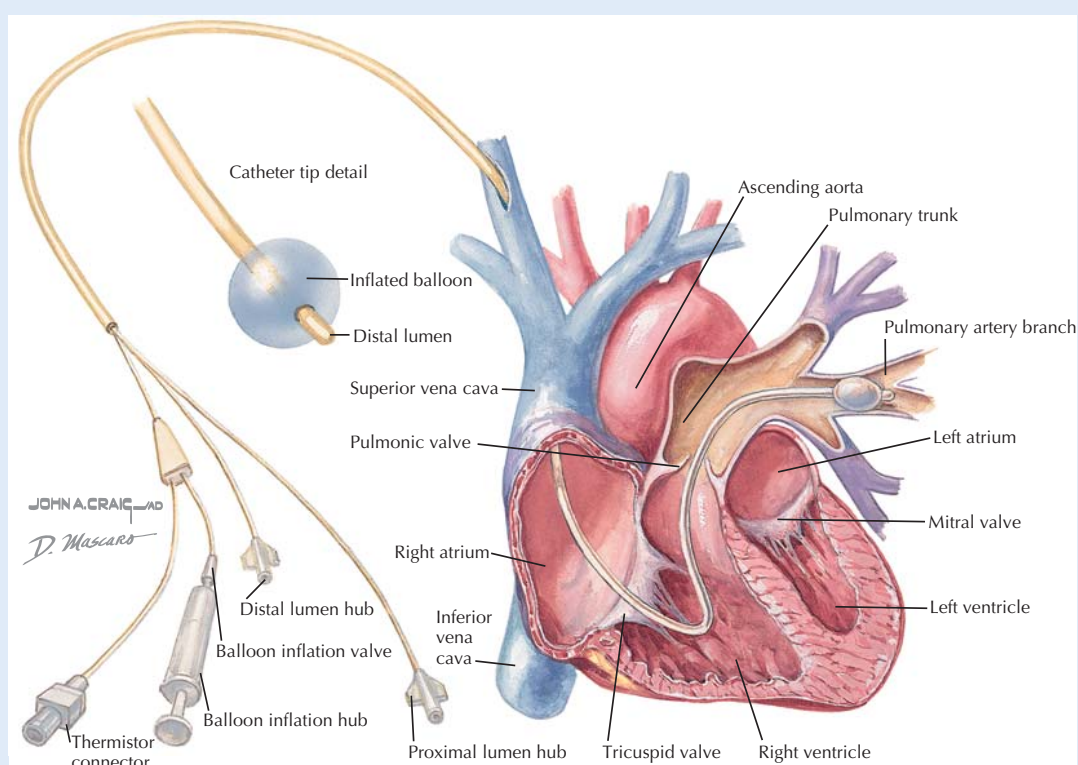
Measurements of Blood Pressure

Arterial pressure is routinely measured by sphygmomanometry, in which a blood pressure cuff is inflated above the systolic pressure, compressing vessels and stopping blood flow. As pressure in the cuff is gradually released, the practitioner listens for the **sounds of Korotkoff** through a stethoscope. These sounds are produced by the pulsatile flow of blood in arteries under the cuff when the cuff pressure is between the systolic and arterial pressure. Thus, the sounds are first heard when cuff pressure falls below systolic pressure, and the sounds disappear when the cuff pressure falls below diastolic pressure. Arterial pressure can also be measured directly through an arterial catheter. Such a catheter can be passed in a retrograde direction (against the flow of blood) for measurement of pressures in the arteries, aorta, and left ventricle, as the catheter advances toward the heart.

Although arterial pressure monitoring is important for detection of hypertension, monitoring of additional pressures is often highly desirable in surgical and intensive care settings. Pulmonary capillary wedge pressure (“wedge pressure”) is one such measurement. A venous catheter can be passed antegrade (in the direction of blood flow) for measurement of pressures in the veins, right atrium, and right ventricle. However, it is not possible to measure

pulmonary venous and left atrial pressures by these direct methods. Instead, we must rely on measurement of pulmonary capillary wedge pressure.

With the aid of a partially inflated balloon, a catheter is passed from a systemic vein, through the right heart, into the pulmonary artery (PA), to a branch of the pulmonary artery. The lumen opens at the distal end of the catheter and is continuous; the balloon is on the outside of the catheter only. When the catheter can be advanced no further, the balloon is fully inflated (the catheter is actually a multilumen Swan–Ganz catheter—one lumen is continuous to the end of the catheter; another leads to the balloon, allowing inflation; other lumens may open along the length of the catheter). Inflating the balloon has no effect on patency of the catheter lumen through which pressure is measured. When the balloon is fully inflated, pressure falls downstream beyond that point. Vascular pressure beyond the point of occlusion equilibrates with downstream pressure, and wedge pressure (measured at the catheter tip) is thus an indicator of pulmonary venous and left atrial pressure. It also approximates left ventricular end-diastolic pressure (LVEDP), when LV pressure has equilibrated with the pressure in the left atrium and pulmonary veins with filling of the ventricle. Wedge pressure is useful in hemodynamic assessment (e.g., in acute heart failure).



Pulmonary Artery Catheterization Pulmonary artery catheters (Swan–Ganz catheters) are multilumen catheters that can be advanced into a branch of the pulmonary artery. With the tip thus wedged with the balloon inflated, flow is occluded and pressure can be measured beyond the balloon occlusion through one of the lumens. The measured *pulmonary capillary wedge pressure* is an approximation of left ventricular end diastolic pressure.

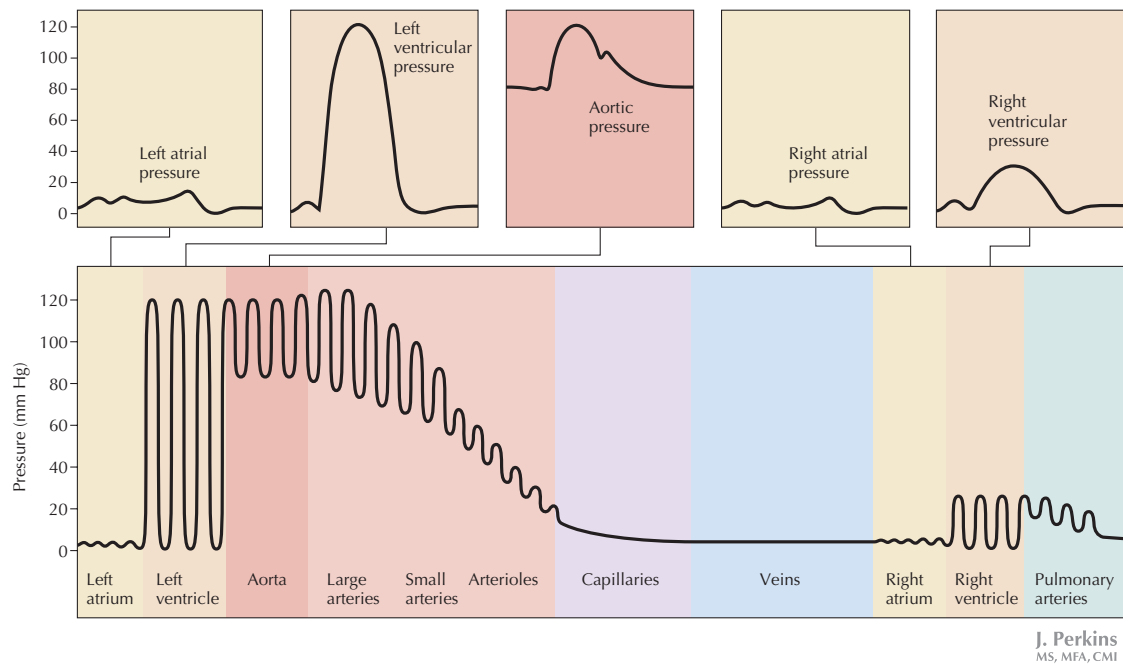


Figure 10.2 Pulse Pressures through the Circulation Changes in pressure pulses are illustrated as blood flows from the left ventricle (LV) to the aorta, through the systemic circulation, and back to the right ventricle (RV) and pulmonary artery (PA).

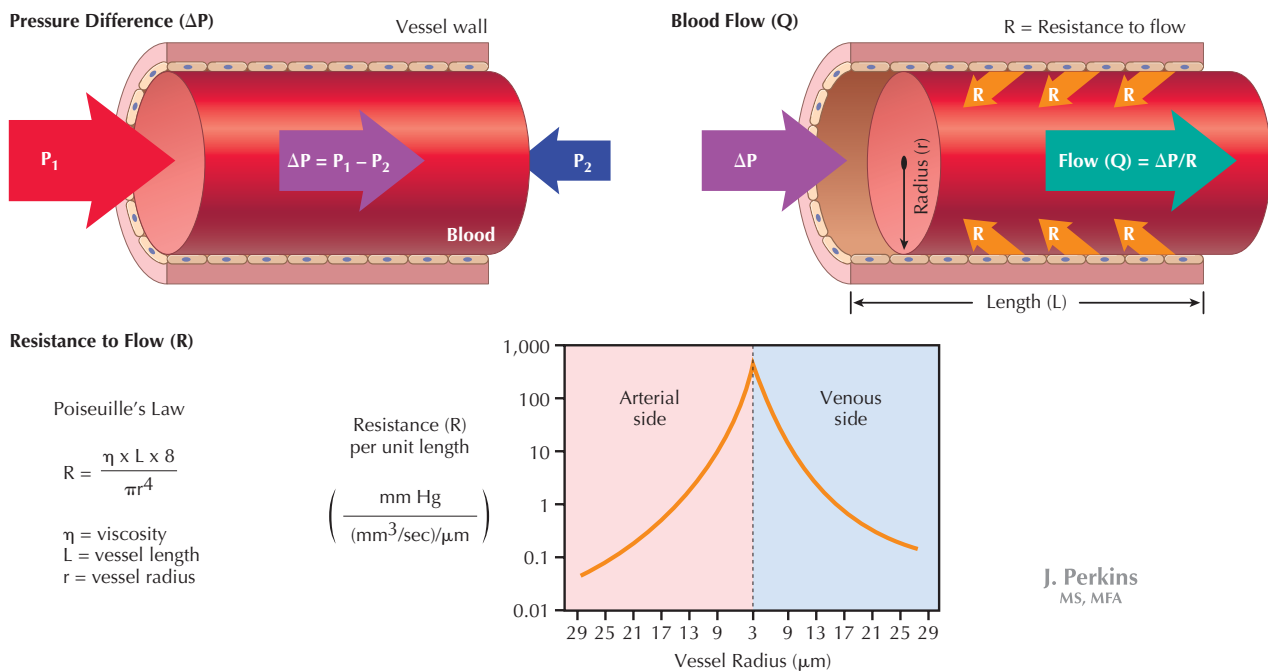


Figure 10.3 Poiseuille's Law Under normal physiological conditions, flow through vessels is governed by Poiseuille's law. By far, the greatest resistance to flow occurs in the smallest vessels constituting the microcirculation, particularly the arterioles. This is due to the small radius of these vessels, and the inverse relationship between resistance and the fourth power of the radius of a vessel.

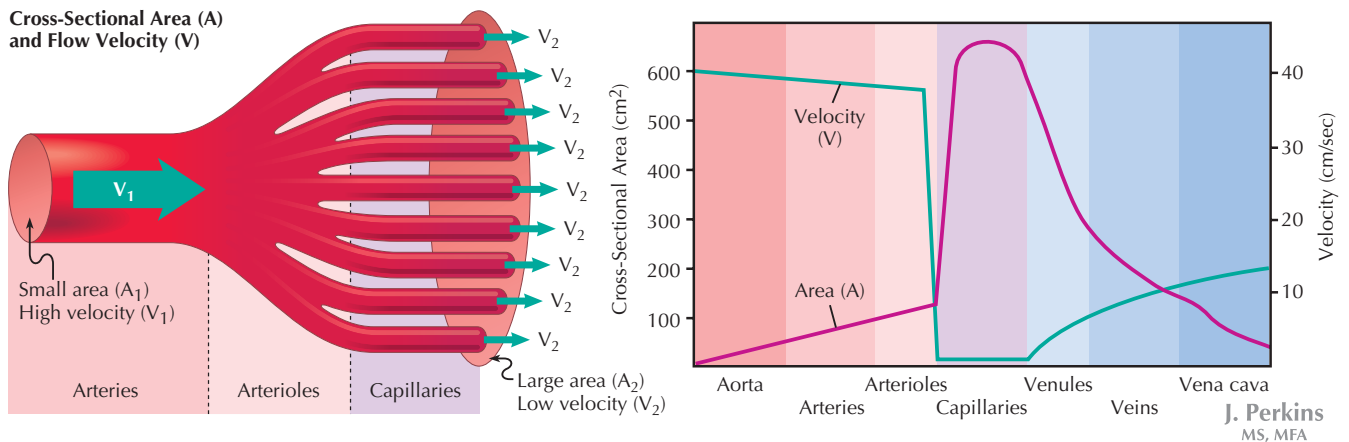


Figure 10.4 Relationship between Velocity of Blood Flow and Cross-Sectional Area Proceeding downstream from the aorta, branching of arterial vessels increases total cross-sectional area and thus results in diminished velocity of blood flow from the aorta to the capillaries. Velocity increases from the capillaries to the large veins with the confluence of vessels and the resulting decrease in total cross-sectional area.

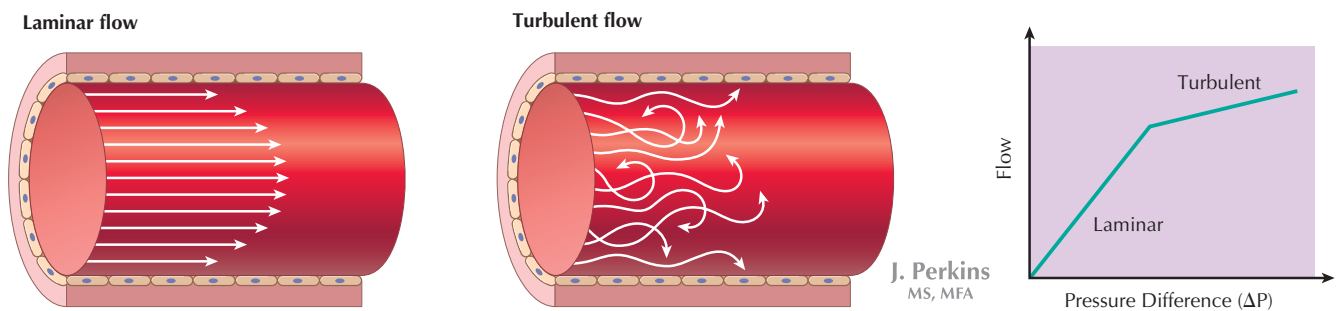


Figure 10.5 Laminar and Turbulent Flow Normally, laminar, or streamlined, flow occurs throughout most of the vascular system. Pathological conditions such as coarctation (narrowing of a vessel), valvular abnormalities, and low blood viscosity (as in anemia) produce turbulence, and are associated with murmurs in the heart or great vessels, or vascular bruits heard by auscultation. Turbulence produces an increase in the pressure gradient required to produce flow.

Key relationships to remember are as follows:

- Resistance is directly proportional to the length of the tube.
- Resistance is directly proportional to the viscosity of the fluid.
- Resistance is inversely proportional to the fourth power of the radius of the tube.

Flow (Q) can also be related to cross-sectional area (A) and linear velocity of the flow (V) in this formula:

$$Q = VA$$

In the cardiovascular system, the overall flow rate (cardiac output) is the same at every level of the circulation, because the circulation is in series. Thus, based on the $Q = VA$ relationship, the velocity of blood flow is slowest in the capillaries, which have the greatest aggregate cross-sectional area

(600 cm²) in the vascular system, and fastest in the aorta, where the cross-sectional area is only about 4 cm² (Fig. 10.4). The low velocity in capillaries is beneficial in terms of exchange of dissolved substances between blood and tissues.

Flow through vessels is generally laminar, with the greatest velocity of flow in the center of the vessel and lowest velocity near the vascular wall (Fig. 10.5). This streamlining of flow is caused by shear stress produced as blood flows past the stationary vessel wall. Turbulent flow, on the other hand, is characterized by irregularities in flow patterns, such as whorls, vortices, and eddies. Vascular disease often is associated with sites of turbulence.

Reynolds' Number (R_e) relates the factors associated with turbulence:

$$R_e = \frac{VD\bar{v}}{\eta}$$

where V is velocity of flow, D is the diameter of the tube, ρ is the density of the fluid, and η is the viscosity of the fluid. When R_e is below 2000, flow is usually laminar; values above 3000 result in turbulence. In the cardiovascular system, density of blood is always close to 1.0 and is not a factor; changes in the other variables are important, along with a number of other factors. Turbulence in blood vessels is associated with audible murmurs and is promoted by:

- high velocity of blood flow
- large vessel diameter
- low viscosity of blood (low hematocrit)
- abrupt changes in vessel diameter, for example in aneurysms (see “Clinical Correlate”) or coarctation (narrowing)
- vascular branch points

Wall tension (T) is another important biophysical parameter in this system and can be conceptualized as the force necessary to hold together a theoretical slit occurring in a vessel wall. This is defined by Laplace’s law:

$$T = P_i r$$

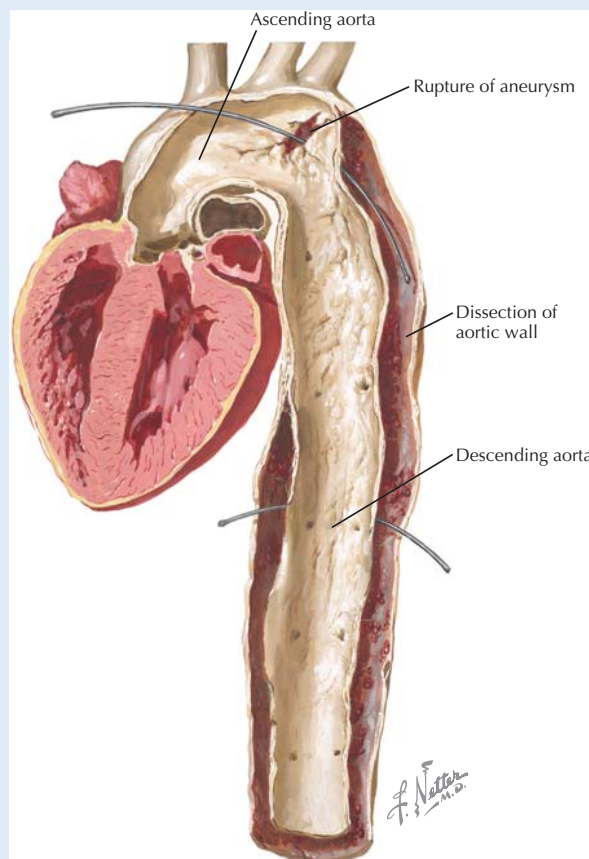
where P_i is the transmural pressure (the difference between pressure inside and outside the vessel, in other words, the pressure gradient across the vascular wall), and r is the vessel radius. Because of the small radii of capillaries and venules, they are protected against rupture despite the transmural pressure gradient. Wall tension is an important consideration in large arteries, where transmural pressures are high and radii are large, especially when the vascular wall is diseased (see “Clinical Correlate”).

CLINICAL CORRELATE

Aortic Aneurysm

An aneurysm is a saclike enlargement in the wall of an artery, caused by weakening of the vessel wall. With bulging, wall stress is further increased, through increased radius of the vessel and decreased wall thickness, according to Laplace’s equation ($T = P_i r$). Risk factors for aortic aneurysm include hypertension, smoking, obesity, atherosclerosis, and hypercholesterolemia.

Rupture of an aortic aneurysm is a medical emergency involving profuse internal hemorrhage and is often fatal. Dissection of an aneurysm is defined as bleeding into the vascular wall through a tear in the inner layer of the vessel (media intima), a common complication in thoracic aortic aneurysms. Aneurysms are treated by replacement of the affected aortic segment by a synthetic graft, or by endovascular implantation of a graft in the area of the aneurysm.



Ruptured and Dissecting Aortic Aneurysm

CHAPTER 11

The Cardiac Pump

Based on the biophysical principles described in Chapter 10, blood flows through the systemic and pulmonary circulations, with the energy for this flow being provided by the pumping action of the heart. Like early mechanical pumps, the heart functions in a cyclic manner, with periods of filling (diastole) and periods of contraction and ejection of blood (systole).

THE CARDIAC CYCLE

The cardiac cycle (or Wiggers diagram) consists of one cycle of ventricular systole and diastole. At rest the cycle is 0.86 seconds in duration if the heart rate is 70 beats per minute. The changes in ventricular pressure and volume, aortic pressure and flow, atrial pressure, venous pulse, electrocardiogram, and phonocardiogram are all interdependent, and understanding the interrelationships between these variables is a key step in comprehending the complexities of hemodynamics (Fig. 11.1).

Left ventricular volume and pressures in the left heart and aorta are illustrated in Figure 11.1. Tracings for the right ventricle and pulmonary artery are similar in shape but substantially lower than those for the left ventricle and aorta. Within this cardiac cycle diagram, there are two short intervals known as periods of **isovolumetric contraction** and **isovolumetric relaxation**. During these periods, all of the valves of the heart are closed. The isovolumetric contraction period for the left heart begins with mitral valve closure (the aortic valve is already closed) and ends with opening of the aortic valve; the isovolumetric relaxation period begins with aortic valve closure and ends when the mitral valve reopens.

There is some slight asynchrony between the left heart and the right heart in terms of valve opening and closure, although the sequences are the same. The sequence of valve opening and closure during the cardiac cycle is as follows:

- *Left heart:* Mitral closure, aortic opening, aortic closing, mitral opening.
- *Right heart:* Tricuspid closure, pulmonic opening, pulmonic closure, tricuspid opening.
- *Aggregate sequence:* Mitral closure, tricuspid closure, pulmonic opening, aortic opening, aortic closure, pulmonic closure, tricuspid opening, mitral opening.



Isovolumetric periods in the cardiac cycle refer to short intervals when all of the valves are closed. On the left side of the heart during isovolumetric contraction, ventricular pressure exceeds atrial pressure, and thus the mitral valve has closed. Ventricular pressure rises as the heart contracts, but it has not yet exceeded aortic pressure, and thus the aortic valve is still closed. During isovolumetric relaxation, ventricular pressure is less than aortic pressure, and thus the aortic valve is closed. However, ventricular pressure has not yet fallen below atrial pressure, and therefore the mitral valve is still closed. The overall sequence in the cardiac cycle in the ventricles consists of isovolumetric contraction, followed by ejection, isovolumetric relaxation, ventricular filling, isovolumetric contraction, and so forth.

The asynchrony between the left- and right-side valves is caused by differences in the pressure gradients between the sides of the circulation, and the degree of asynchrony varies during the respiratory cycle, because of the effects of thoracic pressure on filling and pressures within the heart and central circulation.

Ventricular Pressure

The cardiac cycle is initiated by the P wave of the electrocardiogram (ECG). Depolarization of the atria results in atrial contraction and final filling of the ventricles; this accounts for the slight increase in **left ventricular pressure (LVP)**. When the wave of depolarization reaches the ventricle (QRS complex of the ECG) and ventricular contraction is initiated, LVP rises above **left atrial pressure (LAP)**, producing mitral valve closure and marking the beginning of isovolumetric contraction. **Left ventricular end-diastolic pressure (LVEDP)** is normally about 5 mm Hg (RVEDP is normally about 2 mm Hg). From this point, LVP rises rapidly, as the ventricle attempts to contract against a fixed volume. When LVP rises above aortic pressure, the aortic valve opens, marking the end of isovolumetric contraction and the beginning of ventricular ejection. If aortic pressure is 120/80 mm Hg, this occurs at 80 mm Hg. LVP continues to rise for a period, but as the stroke volume is ejected, LVP reaches a peak (at approximately 120 mm Hg when aortic pressure is 120/80 mm Hg) and begins to fall. The T wave of the ECG occurs at this time,

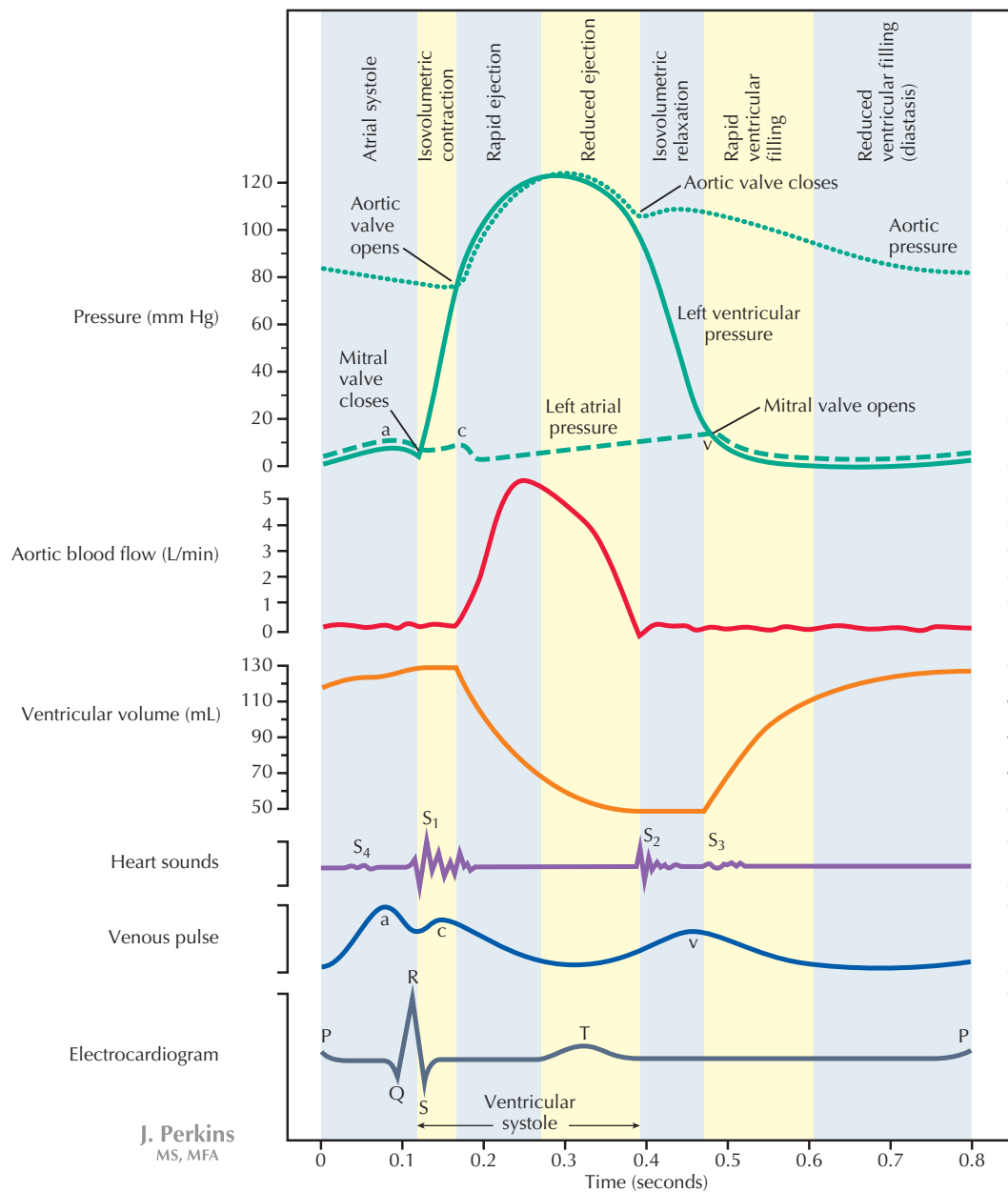


Figure 11.1 The Cardiac Cycle The cardiac cycle (or Wiggers diagram) is a simultaneous depiction of several parameters related to blood flow and volume, the electrocardiogram, and the echocardiogram through a cycle of cardiac systole and diastole. Ventricular systole begins with a short period of isovolumetric contraction, during which all of the heart valves are closed. This is followed by the ejection phase of systole, and then isovolumetric relaxation when the heart valves are again all closed. Cardiac filling occurs after isovolumetric relaxation. The temporal relationships between features of these various curves or tracings are predictable, based on functional relationships within the cycle. The a, c, and v waves are identified on both the venous pulse and atrial pressure curves.

as ventricular repolarization takes place. LVP falls below aortic pressure, but aortic valve closure is not immediate, due to the momentum associated with ventricular ejection. Closure of the aortic valve marks the end of ejection and the beginning of isovolumetric relaxation. During this period, the ventricle

is beginning to relax, but the valves are closed, resulting in rapid fall of LVP. When LVP falls below LAP, the mitral valve opens, and ventricular filling begins. During the remainder of diastole, LVP remains slightly below LAP, consistent with the direction of blood flow during ventricular filling.

Aortic Pressure

The **aortic pressure** curve falls during diastole as blood flows to the peripheral circulation (see Fig. 11.1). The upstroke in this curve begins (at 80 mm Hg if aortic pressure is 120/80 mm Hg) with the opening of the aortic valve. As ejection proceeds, pressure rises to the systolic aortic pressure (120 mm Hg in this example), and then begins to fall. The **dicrotic notch**, a high frequency deflection in the aortic pressure curve, occurs when the aortic valve closes. **Aortic flow** reaches a peak during rapid ejection and is lowest during diastole when no ejection is occurring.

Ventricular Volume

The **left ventricular volume** curve illustrates the filling and emptying of the ventricle (see Fig. 11.1). At the time of the P wave, ventricular filling is nearly complete, but the final filling (approximately 15%) is produced by atrial contraction. This phase of ventricular filling is known as **active ventricular filling**. Ventricular filling ends when the mitral valve closes, at the left ventricular **end-diastolic volume** (EDV). Of course, ventricular volume remains constant during isovolumetric contraction. With opening of the aortic valve, rapid ejection of blood begins, and ventricular volume falls quickly. This is followed by a period of reduced ejection, which ends with aortic valve closure. At this point, left ventricular **end-systolic volume** (ESV) has been reached, and this volume remains constant during isovolumetric relaxation. The stroke volume is the difference between EDV and ESV. Normal EDV is approximately 140 mL and normal stroke volume is approximately 70 mL, although these values vary considerably. Opening of the mitral valve initiates a period of **rapid passive filling**. This is followed by a period of **slow passive filling** (also known as **diastasis**), as the ventricle fills more slowly with the diminished pressure gradient between the atrium and ventricle.

Atrial Pressure

The shape of the **left atrial pressure** curve reflects both cardiac events and venous return (flow of blood back to the atrium); it consists of the a, c, and v waves. Following the P wave on

the ECG, atrial contraction produces the first rise in atrial pressure, the a wave (see Fig. 11.1). During isovolumetric contraction, there is another upward wave in the LAP curve, the c wave, caused by bulging of the mitral valve back into the left atrium, as the ventricle attempts to contract against a fixed volume. During the ejection phase of the left ventricle, LAP rises slowly while venous return from the pulmonary circulation fills the atrium (the mitral valve is closed during this period), producing the v wave. When LVP falls below LAP, the mitral valve opens and LAP falls as blood flows into the left atrium. For the remainder of diastole, LAP remains above LVP as the heart fills. Because there are no valves between the vena cavae and right atrium (or the pulmonary vein and left atrium), the **venous pulse** and the atrial pressure curve are similar in shape.

Phonocardiogram

The **phonocardiogram** is an acoustical recording reflecting the **heart sounds** generated during the cardiac cycle (see Fig. 11.1). S_1 , the first heart sound, is generated by closure of the mitral and tricuspid valves (the mitral valve closes slightly before the tricuspid valve). S_2 is associated with closure of the aortic and pulmonic valves; again, the valve on the left side closes just before the right-side valve. During inspiration, closure of the pulmonic valve is delayed more distinctly, normally resulting in audible **physiological splitting** of S_2 . This delay is caused by increased filling of the right ventricle, due to reduced intrathoracic pressure during inspiration. S_3 , the third heart sound, is normal in children, and is associated with rapid ventricular filling. It is not heard in normal adults, and may be a sign of volume overload, as in congestive heart failure. S_4 , caused by active ventricular filling, also is not audible in normal adults, but may be present in a variety of disease states.

REGULATION OF CARDIAC OUTPUT

Cardiac output is 5 liters (L) per minute (min) in the average, resting adult. The **cardiac output formula** is a key physiological concept, relating cardiac output (CO) to heart rate (HR) and stroke volume (SV):

$$CO = HR \times SV$$

At rest, HR is approximately 70 beats per minute and SV is approximately 70 mL, yielding CO of approximately 5000 mL/min (5 L/min). In order to maintain adequate arterial blood pressure and blood flow at rest, during increased activity, or during challenges to homeostasis such as exercise, CO must be adjusted through regulation of HR and SV. Many of the mechanisms that regulate HR or SV affect both parameters, and furthermore, are closely linked to mechanisms involved in blood pressure regulation.

The **autonomic nervous system** (ANS) plays an important role in regulation of cardiac output, through modulation of



The events of the cardiac cycle are initiated when an action potential is generated in the sinoatrial (SA) node, producing depolarization and contraction of the atria, followed by contraction of the ventricles. When heart rate is increased, diastole is shortened, because the P wave (and thus, atrial and ventricular contraction) occurs earlier. As a result, the time for ventricular filling is diminished. However, at a resting heart rate of 70 beats per minute, there is a significant period of diastasis (slow ventricular filling), and, up to a point, increasing heart rate will not limit the time for rapid passive filling, the period during which most ventricular filling occurs. Very rapid heart rates allow insufficient time for ventricular filling and can even compromise cardiac output.

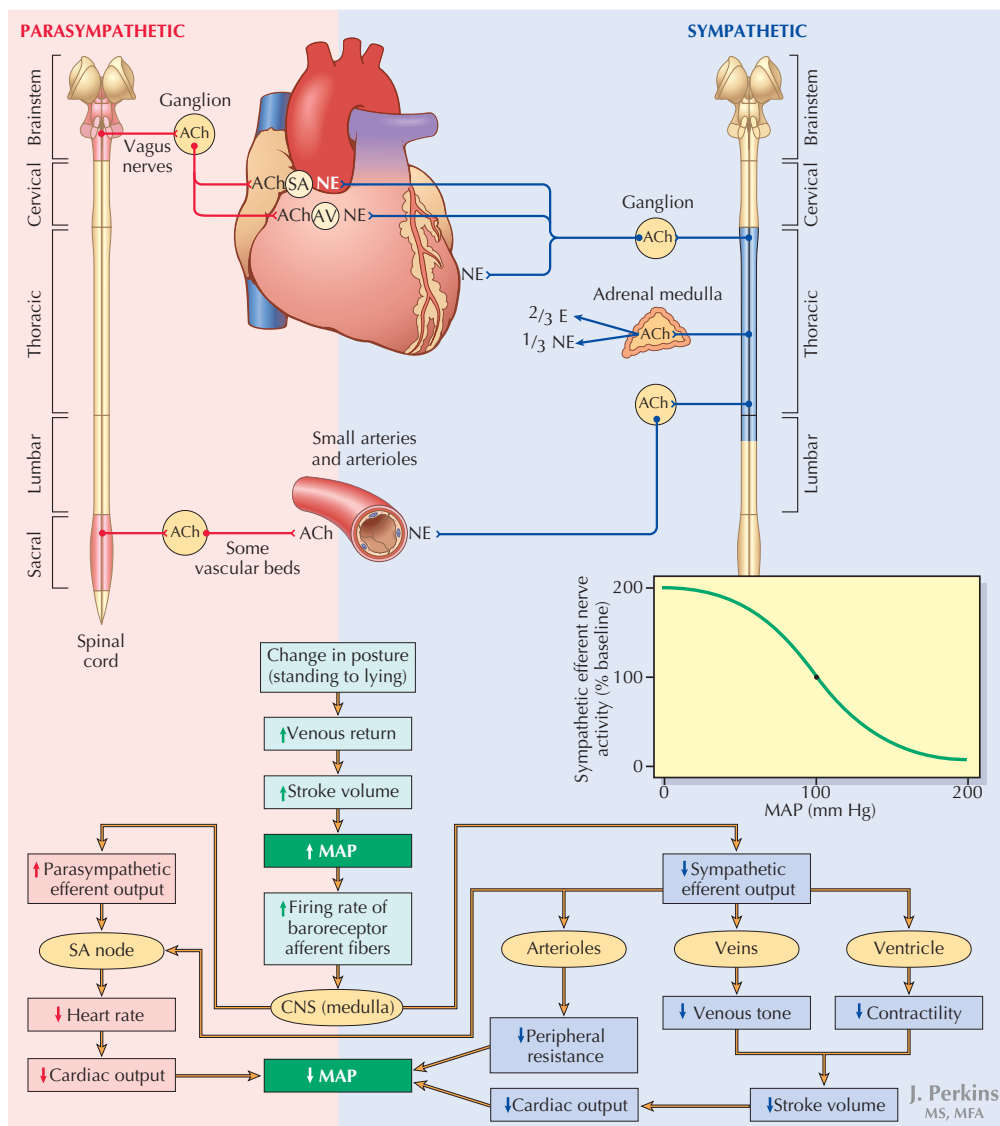


Figure 11.2 Autonomic Nervous System and Arterial Baroreceptor Reflexes Both sympathetic and parasympathetic nerves innervate the SA and AV nodes. The myocardium is innervated by sympathetic nerves. Arterial and venous vessels throughout most of the body are innervated by sympathetic nerves, whereas the parasympathetic nervous system innervates vessels in the genital organs and lower GI tract. Autonomic efferent activity is regulated by the baroreceptor reflex, in response to changes in arterial pressure detected at the carotid sinus and aortic arch baroreceptors. The response to a change in posture is illustrated. ACh, acetylcholine; E, epinephrine; MAP, mean arterial pressure; NE, norepinephrine.

the SA nodal pacemaker rate, myocardial contractility, and vascular smooth muscle tone (Fig. 11.2). Sympathetic nerve endings at the heart and vessels release norepinephrine, while parasympathetic nerves release acetylcholine at the heart (primarily the SA node) and in a few vascular beds (vessels in most regions do not have parasympathetic innervation). In addition to direct release of catecholamines by sympathetic nerves in the heart and vessels, circulating catecholamines (predominantly epinephrine) released by the adrenal gland

contribute to cardiovascular regulation when the sympathetic nervous system is strongly activated.

Heart Rate Regulation

The arterial **baroreceptor reflex** is a key mechanism involved in autonomic regulation of cardiovascular function (see Fig. 11.2). The baroreceptors consist of specialized cells in the vascular wall of the carotid sinuses and aortic arch. Barorecep-

tors respond to the stretch associated with elevation of arterial pressure by increasing afferent impulses to the medullary cardiovascular center, where autonomic nervous system activity is regulated. In response to increased afferent input, sympathetic nerve activity is reduced, and parasympathetic outflow is increased. The effects of these autonomic responses on the heart and vessels result in a return of arterial blood pressure toward its original level; specific aspects of this regulation are discussed in the following text.

Mechanisms that contribute to **regulation of heart rate** include the following:

- The autonomic nervous system and baroreceptors.
- The Bainbridge reflex response to atrial stretch.
- Effects of thoracic pressure changes during respiration on venous return.

Under normal conditions, heart rate regulation is accomplished primarily by the autonomic nervous system and baroreceptor reflexes. Sympathetic nerves release norepinephrine at the SA node, which acts at β_1 receptors, increasing cAMP production and ultimately raising the activity of the pacemaker and elevating heart rate. Parasympathetic activation reduces heart rate through release of the neurotransmitter acetylcholine. Thus, when arterial pressure rises, the baroreceptor reflex results in a fall in the SA node pacemaker rate. The resulting reduction in heart rate contributes to a fall in CO, and a return to normal arterial pressure (see Fig. 11.2). Conversely, reduced arterial pressure will result in less stretch of baroreceptors, reduced afferent impulses to the cardiovascular center, greater sympathetic activity, and reduced parasympathetic activity, raising heart rate and returning arterial blood pressure to the normal level.

Another reflex potentially regulating heart rate is the **Bainbridge reflex**. When right atrial volume is increased, low-pressure stretch receptors (that is, receptors responding to stretch at the low pressures typical in the atria) initiate a neural reflex that increases heart rate through sympathetic nerves. Note that arterial baroreceptors respond to stretch by decreasing heart rate, whereas atrial baroreceptors respond to stretch by increasing heart rate. In the former case, the response is part of a mechanism to regulate arterial pressure, and in the latter case, the response is to increased blood volume. When atrial stretch results in increased HR, blood volume will be redistributed. In addition, a number of other mechanisms for regulating blood volume and pressure will be activated; these will be considered subsequently (see “Regulation of Arterial Blood Pressure” in Chapter 12).

Heart rate may also be affected by the respiratory cycle, particularly in infants and children. In so-called **respiratory sinus arrhythmia**, heart rate is increased during inspiration and decreased during expiration. The low-pressure baroreceptors within the atria are stretched by increased venous return during inspiration, producing the cyclic variation in HR.



The Bainbridge reflex and the arterial baroreceptor reflex produce opposite responses to an intravenous infusion. The Bainbridge reflex is usually observed when a rapid intravenous infusion is administered to a subject with a slow heart rate. Stretch of atrial receptors produces a reflexive increase in heart rate under these circumstances. On the other hand, when heart rate is high to begin with, for example, after hemorrhage, intravenous infusion usually decreases heart rate. In this circumstance, the increased ventricular filling associated with the infusion produces an increase in cardiac output and therefore an increase in arterial pressure. The rise in arterial pressure causes stretch of arterial baroreceptors, resulting in diminished sympathetic nervous system activity and enhanced parasympathetic activity, and a slowing of heart rate.

Stroke Volume Regulation

Stroke volume regulation is dependent on several parameters:

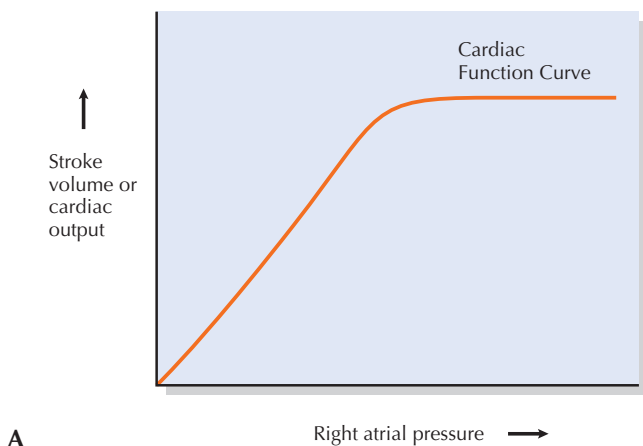
- **Preload:** The degree of stretch of myocardial fibers prior to contraction. Preload in the intact heart is closely related to end-diastolic volume of the ventricle. Increased preload is associated with increased force of contraction, and hence, increased stroke volume.
- **Afterload:** The force against which the heart contracts. In the intact heart, afterload is closely related to arterial pressure or left ventricular pressure during systole. Increased afterload opposes ejection by the heart and therefore will tend to reduce stroke volume.
- **Contractility (inotropism):** The intrinsic ability of cardiac muscle to generate force at a given fiber length. Specifically, contractility is *not* synonymous with force of contraction, because force of contraction is dependent on degree of initial stretch of the muscle fibers (preload).

The Frank-Starling Relationship

The **Frank-Starling relationship** is named for Otto Frank, who described the relationship between pressure generated in the frog heart and the diastolic volume, and for Ernest Starling, who demonstrated, in an intact, canine heart-lung preparation, the relationship between stroke volume and end-diastolic volume. The **cardiac function curve** illustrates this relationship (Fig. 11.3A). When preload is increased by raising atrial pressure, stroke volume (and therefore, cardiac output) rises, up to the optimum level of preload. This is an intrinsic mechanism for regulation of stroke volume and is not associated with a change in inotropism. The molecular basis for this relationship involves stretch of sarcomeres; at optimal preload, conditions for subsequent crossbridge formation and recycling are best.

Sympathetic Regulation of Stroke Volume

Stroke volume is also regulated by the **sympathetic nervous system** (Fig. 11.3B). Cardiac muscle is directly innervated by



A

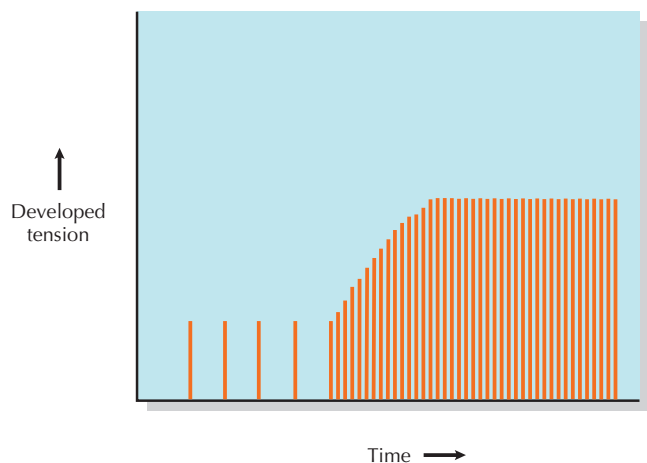
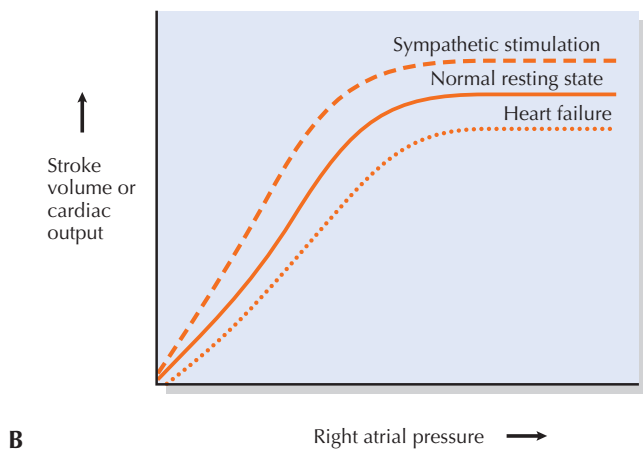


Figure 11.4 Effect of Heart Rate on Force of Contraction (Treppe or Staircase Effect) When the intervals between cardiac muscle contractions are long, the tension developed is low. Note the staircase-like increase in the force of contraction when frequency (heart rate) is increased. Because this effect is not dependent on changes in resting fiber length, contractility (inotropism) is increased.



B

Figure 11.3 The Frank-Starling Relationship A, An increase in preload of the ventricle results in an increase in stroke volume and thus cardiac output. B, Sympathetic stimulation of the heart shifts this curve upward and to the left; heart failure results in a lower slope of the curve. The slope of the cardiac function curve is one measure of contractility of the heart. In this graph, it is assumed that afterload (arterial pressure) is held constant.



The Frank-Starling relationship is an important mechanism for matching cardiac output and venous return, and left- and right-side cardiac output. If output from the right ventricle is suddenly increased (for example, due to a change in posture), return to the left ventricle will be rapidly elevated. The resulting stretch of the left ventricle will cause an increase in its output, matching the rise in output that occurred on the right side. Thus, the Frank-Starling relationship maintains balance between right- and left-side output. It is also an important mechanism in the adjustment of stroke volume and cardiac output during the complex physiological changes associated with exercise or volume depletion (for example, in hemorrhage).

sympathetic nerves, which release norepinephrine. Norepinephrine, through binding to β -adrenergic receptors (β_1), produces elevation of intracellular free Ca^{2+} levels, resulting in increased contractility of the heart. This is a true increase in inotropism or contractility, because it is independent of any change in preload. Release of epinephrine by the adrenal medulla may also produce greater contractility of the heart, although levels of circulating epinephrine rarely reach levels that significantly affect inotropic activity. The parasympathetic nervous system has only limited effects on contractility of human ventricles. Clinically, a number of drugs are used to promote inotropism in heart failure patients, including digitalis, dopamine, and dobutamine. Although the basic cardiac function curve simply describes the effects of preload on cardiac output or stroke volume, the slope of this curve is a measure of inotropism. With sympathetic stimulation, the slope is steeper, indicating enhanced contractility.

Treppe, also known as the **staircase effect**, is an intrinsic mechanism for regulation of stroke volume. When cardiac contractions are infrequent, force of contraction is reduced; as heart rate increases, force of contraction is elevated (Fig. 11.4). Because these changes occur independently of changes in preload, they reflect a change in contractility. The increased contractility at higher heart rates is associated with elevated free intracellular Ca^{2+} in myocardial fibers.

During a baroreceptor-mediated response to a fall in arterial pressure (see Fig. 11.2), many mechanisms contribute to the change in cardiac output. Sympathetic activation elevates heart rate, as discussed previously, but unless mechanisms are available for maintaining or elevating stroke volume simulta-

neously, elevation of heart rate is not an effective mechanism for raising cardiac output, because filling time for the heart is reduced. By raising contractility of the heart, sympathetic activation enhances stroke volume as well, and, simultaneously, sympathetic constriction of the systemic venous system elevates preload of the heart, allowing the Frank-Starling relationship to further augment stroke volume and cardiac output.

ASSESSMENT OF CARDIAC FUNCTION

Force–Velocity Curves

Assessment of cardiac function and, in particular, inotropism is important both clinically, in heart failure patients, and experimentally. In addition to analysis of the **cardiac function curve** and its slope, analysis of the **force–velocity relationship** in cardiac muscle illustrates the effects of preload and altered contractility on cardiac function (Fig. 11.5). Experimentally, force–velocity curves are usually constructed by measuring velocity of contraction of isolated segments of cardiac muscle against various afterloads. When velocity of contraction is plotted against afterload, an inverse relationship is revealed: when the force of contraction is higher (i.e., when the muscle contracts against higher afterload), velocity of shortening is reduced. Maximum force of contraction occurs at zero velocity, that is, during an isometric contraction. On the other hand, velocity is highest (V_m) at zero afterload. Changes in preload produce a family of curves, all with the same y-intercept (the same maximum velocity of contraction at zero afterload). These curves are another manifestation of the Frank-Starling relationship: greater preload generally results in greater force of contraction, and thus velocity of contraction, although V_m is unchanged. V_m is a measure of the contractile state of the tissue, representing the maximal rate at which actin and myosin filaments can interact to produce contraction. In contrast to the effects of changes in preload, positive inotropic influences such as sympathetic stimulation change V_m . The force–velocity curve is shifted upward and to the right, with an increase in the maximal force that can be developed by the muscle (during an isometric contraction, at the x-intercept) and an increase in V_m (at the y-intercept). An increase in V_m is indicative of a positive inotropic effect.

Pressure–Volume Relationship

The ventricular **pressure–volume loop** is a continuous measurement of ventricular volume against ventricular pressure during the cardiac cycle (Fig. 11.6). During the period of isovolumetric contraction, volume is constant, but pressure rises rapidly. During the ejection phase of the cycle, volume falls, while pressure remains high. Subsequently, during isovolumetric relaxation, volume is constant and pressure falls rapidly. This cycle is then repeated over and over, with each

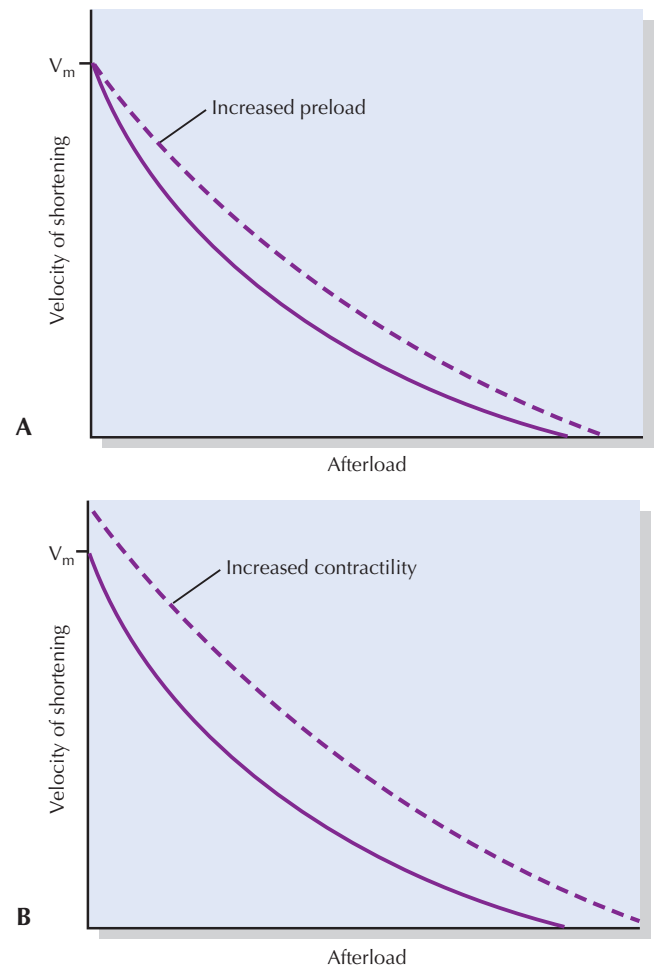


Figure 11.5 Force–Velocity Relationship in Cardiac Muscle Force–velocity curves depict the effect of afterload on velocity of contraction of cardiac muscle. *A*, When the force of contraction is higher (i.e., when the muscle contracts against higher afterload), velocity of shortening is reduced. Velocity is highest (V_m) at zero afterload. Increased preload shifts the curve upward, but V_m , a measure of contractility, is unchanged. *B*, Positive inotropic influences such as sympathetic stimulation change V_m . The force–velocity curve is shifted upward.

beat. The shape and the area within this loop are affected by changes in preload, afterload, and contractility.

Ejection Fraction

A simple and useful measurement for assessment of myocardial function is the **ejection fraction**. Ejection fraction is the ratio of stroke volume to end-diastolic volume. In a healthy person at rest, this should be greater than 50%. If a positive inotropic drug is administered, ejection fraction will rise; in myocardial ischemia or heart failure, ejection fraction is diminished. Ejection fraction can be estimated noninvasively by echocardiography.

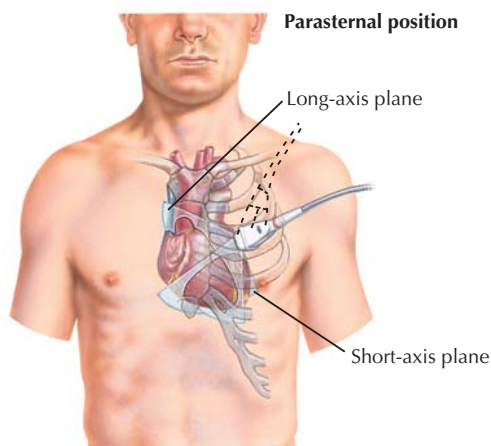
CLINICAL CORRELATE

Echocardiography

Echocardiography is a technique in which inaudible, high-frequency sound is transmitted through the thorax (transthoracic echocardiography) or esophageal wall (transesophageal echocardiography) into the heart. The echoes of the ultrasound, reflected at the interfaces of tissue and fluid, are recorded to produce a graphic image of the heart. In standard echocardiography, the resulting echocardiogram is a two-dimensional image of a slice of

the heart. In Doppler echocardiography, the use of continuous wave Doppler ultrasound allows assessment of the velocity of blood flow within the heart. Using these techniques, it is possible to determine size and structure of valves and chambers; to determine ejection fraction and other parameters; and to detect abnormal wall motion, valvular regurgitation (leakage) or narrowing, bacterial growths (vegetation) within the heart, and other pathologies.

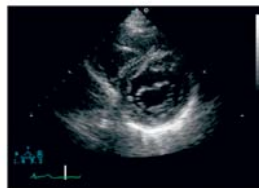
Transducer Positions in Echocardiographic Examination



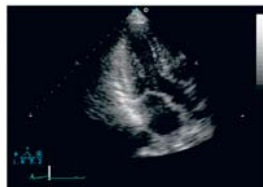
Left parasternal position allows views in long- and short-axis planes. Tilting transducer allows multiple sections.



Normal long-axis view during systole



Normal short-axis view at mitral valve level

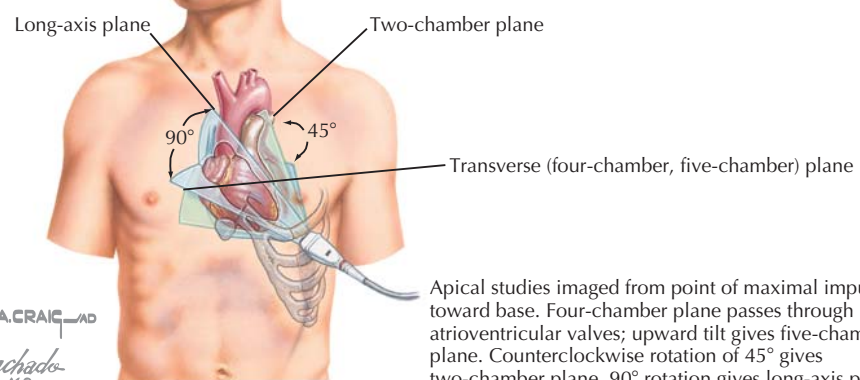


Normal apical long-axis view



Normal apical four-chamber view

Apical position



Apical studies imaged from point of maximal impulse toward base. Four-chamber plane passes through atrioventricular valves; upward tilt gives five-chamber plane. Counterclockwise rotation of 45° gives two-chamber plane. 90° rotation gives long-axis plane.

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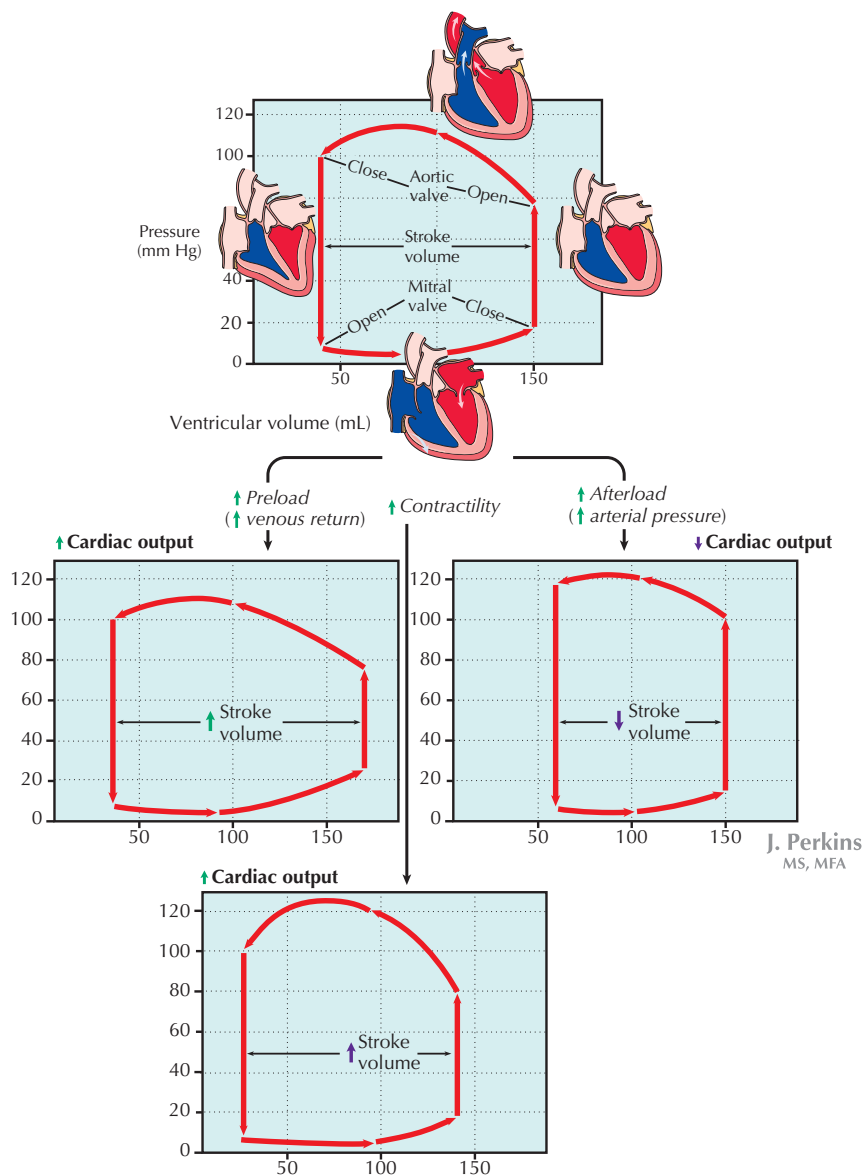


Figure 11.6 Ventricular Pressure-Volume Loop A continuous plot of ventricular volume and ventricular pressure during the cardiac cycle yields a closed loop. Following the *red arrows* from the bottom left of the top graph, during diastole, filling of the ventricle occurs with little change in pressure; during isovolumetric contraction, volume is constant, but pressure rises rapidly. During the ejection phase of the cycle, volume falls, while pressure remains high. During isovolumetric relaxation, volume is constant and pressure falls rapidly, to the starting point on the diagram. Changes in preload, afterload, and contractility affect stroke volume as illustrated.

VASCULAR FUNCTION AND CARDIAC OUTPUT

Regulation of cardiac output depends not only on adjustment of heart rate and stroke volume, but also on regulation of the function of veins and venous return. Venous return is defined as the flow returning to one side of the heart. Thus, since cardiac output is defined as flow from one ventricle, the normal value for venous return will be identical to cardiac

output when averaged over time. If cardiac output is 5 L/min, venous return will also be 5 L/min. Matching of cardiac output and venous return is an important aspect of cardiovascular function.

Just as cardiac output is affected by various factors, so is venous return. Flow from the capillaries to the right atrium occurs despite a small pressure gradient. Because this

pressure gradient is small, venous return is affected by factors that are often different from those affecting the arterial system.



Cardiac output and venous return can be conceptualized as two sides of the same coin, cardiac output being the flow from one side of the heart and venous return being the flow back to one side of the heart. Measured over time, cardiac output and venous return must be equal, since cardiac output is completely dependent upon return of blood to the heart (venous return). A change in one of these parameters will produce an equivalent change in the other.

Venous Compliance and Gravitational Effects

An important factor affecting veins is **compliance** (the change in volume associated with a change in pressure). Compliance of veins is approximately 20-fold greater than that of arteries. Thus, when a person stands from a recumbent position, blood pools in the lower extremities as veins distend due to the hydrostatic pressure associated with the column of blood between the lower extremities and the right atrium. Cardiac output is transiently reduced upon standing as pooling occurs. To prevent **orthostatic hypotension** (decrease in blood pressure upon standing) and to restore venous return under these conditions, the most important compensatory mechanism is the **baroreceptor reflex** (see Fig. 11.2). Increased sympathetic outflow results not only in constriction of the arteries and augmentation of rate and contractile force of the heart, but also constricts veins, reducing the pooling effects of gravity. Veins, in fact, act as a reservoir of blood, which can be mobilized by sympathetic nervous system activation and venoconstriction, to increase venous return and, thus, cardiac output. This is an important mechanism in maintaining cardiac output during volume depletion as well. Movement of skeletal muscles is also important in preventing orthostatic hypotension. During walking, for example, movement of leg muscles compresses veins and augments venous return, because veins outside the central venous system contain one-way valves that prevent backflow of blood.

Effect of Respiration

Another factor affecting venous return is the **effect of respiration**. During inspiration, as the rib cage expands and the diaphragm moves down, negative pressure is created in the thorax. Simultaneously, pressure in the abdominal cavity rises, due to the downward movement of the diaphragm. Thus, veins in the abdomen are subjected to a positive pressure, which augments venous return toward the negative pressure in the thorax. During expiration, the gradient is reduced. With increased depth and frequency of respiration, for example during exercise, there is a pulsatile increase in venous return.

Cardiac Function and Vascular Function Curves

The interactions between vascular function and cardiac function can be illustrated by the simultaneous consideration of two relationships: the **cardiac function curve** (previously considered) and the **vascular function curve** (Fig. 11.7). The cardiac function curve illustrates the Starling relationship, in which right atrial pressure is the independent variable and cardiac output is the dependent variable. A rise in right atrial pressure (preload) produces a rise in cardiac output. The vascular function curve as illustrated is an unconventional graph, in which the independent variable (cardiac output) is plotted on the y-axis, and the dependent variable (right atrial pressure) is on the x-axis. This is an inverse relationship: a rise in cardiac output produces a fall in right atrial pressure (or preload). In other words, greater cardiac output will result in redistribution of blood volume, with reduction of preload. Note that the x intercept of the vascular function curve is the **mean circulatory pressure**. This is the pressure in the system when cardiac output is zero, and is dependent on blood volume and the compliance of the vascular system as a whole. Thus, if the heart is stopped, pressure equilibrates across the entire cardiovascular system. A positive mean circulatory pressure is necessary for the heart to effectively pump blood.



Mean circulatory pressure is the residual pressure that would exist throughout the cardiovascular system (after equilibration) if the heart were suddenly stopped and vascular tone throughout the system were to remain the same. The normal value for mean circulatory pressure is about 7 mm Hg and is a function of vascular tone (particularly venous tone) and blood volume. Without positive mean circulatory pressure, efficient circulation of blood would not be possible, because preload would be compromised at even a very low cardiac output.

Integration of the Cardiac and Vascular Function Curves

When the cardiac function and vascular function curves are plotted on the same graph, the two curves intersect at one point, the resting cardiac output and right atrial pressure (normally about 5 L/min and 2 mm Hg); in other words, this is the steady-state or equilibrium point for normal resting cardiovascular function. When one of the curves is altered, a new equilibrium point is reached. For example, an increase in blood volume (hypervolemia) will shift the vascular function curve upward and to the right (Fig. 11.7B). At the new equilibrium point for the two curves (point B), cardiac output is increased, due to higher preload. Venous constriction would have the same effect as increased volume. On the other hand, hypovolemia will shift the vascular function curve to the left and downward, and the new equilibrium point A represents lower cardiac output at rest. The cardiac mechanism resulting

venous tone is unchanged. Thus, if TPR is diminished, the vascular function curve rotates up and to the right, because venous and right atrial pressure will be raised as a result of the lower arterial resistance. Note that mean circulatory pressure is unchanged, because total compliance of the vascular system is not significantly affected. The cardiac function curve shifts upward, as stroke volume is ejected against lower arterial pressure (reduced afterload) due to reduced TPR. Thus, the equilibrium point for the system moves from point X to point B, at which cardiac output is elevated from the original steady state. Opposite changes occur when TPR is elevated. The vascular function curve rotates downward and to the left, and the cardiac function curve shifts downward, due to the effects of increased afterload. The equilibrium shifts from point X to point A, at which cardiac output is reduced from the original steady state.

CLINICAL CORRELATE

Venous Pressure in Heart Failure

In right-heart failure, contractility of the right ventricle is reduced, often due to myocardial infarction. As a result, cardiac output will be reduced and venous pressure will be elevated. This knowledge can be useful diagnostically. If a normal individual is lying in bed with his upper body tilted up 30 to 60 degrees on a pillow, the external jugular vein is ordinarily collapsed just above the level of the clavicle. The height of the column of blood in the external jugular vein is a measure of central venous pressure or right atrial pressure. If a patient is in heart failure and the central venous pressure is elevated, the vein will be distended significantly above the clavicle.

CHAPTER 12

The Peripheral Circulation

The peripheral circulation consists of the systemic arteries, veins, and microcirculation. The structures of arteries and veins include three tissue layers (Fig. 12.1):

- **Tunica intima:** This innermost layer consists of a single endothelial cell layer forming the inner lining of the vessel; these cells rest on a basement membrane that separates the intima from the media.
- **Tunica media:** The media consists mainly of smooth muscle and is the contractile portion of the vascular wall.
- **Tunica adventitia:** The adventitia consists mainly of connective tissue.

There is variation in the absolute and relative thickness of media and adventitia between arteries and veins, as well as between small and large vessels; differences are also observed between vessel types in the connective tissue and cellular constituents of these layers. For example, the walls of large arterial vessels are rich in elastic tissue and have a relatively thick adventitia compared with smaller arteries. On the other hand, smaller arteries have a relatively more dominant, muscular medial layer. Capillaries, unlike other vessels, have no media or adventitia. Their vascular walls consist simply of endothelial cells and basement membrane.

THE MICROCIRCULATION AND LYMPHATICS

The microcirculation consists of vessels less than 100 micrometers (μm) in diameter and includes arterioles, metarterioles, capillaries, and venules (Fig. 12.2). Muscular arterioles may directly feed into capillaries or into metarterioles that structurally are between capillaries and arterioles. Precapillary sphincters are bands of smooth muscle found at the point at which blood enters capillaries. Constriction and relaxation of the smallest arteries, arterioles, and precapillary sphincters regulates flow into capillary beds (see “Regulation of Blood Flow”).

The thin wall of capillaries consists of a single layer of endothelial cells and the associated basement membrane. This simple structure is well adapted for the diffusion of gases, nutrients, and wastes between blood and the interstitial fluid of tissues, which occurs only in capillaries. The exchange of



The histological differences between various types of vessels reflect their functional roles. The thin wall of the capillaries, consisting of only an intimal layer, allows the efficient exchange of nutrients, waste, and dissolved gases between blood and tissues. On a relative basis, the muscular tunica media is more prominent in small arteries and arterioles than in large arteries or veins, reflecting their role in regulation of blood flow. Larger arteries have thick adventitia with significant elastic tissue, consistent with their role as “distributing vessels,” whereas veins have more compliant adventitia. The tunica media of veins contains smooth muscle arranged both circularly and longitudinally, reflecting the function of veins as “capacitance vessels.” Similarly, elastic tissue in the adventitia of veins is less prominent than in arteries.

fluid between the vascular space and interstitial space also occurs across the capillary wall by simple diffusion (see Fig. 2.1). The net filtration pressure for diffusion of fluid out of capillaries is governed by the Starling equation. Entering the capillary at the arteriolar end, hydrostatic pressure is approximately 30 mm Hg; it falls to approximately 10 mm Hg at the venular end of the capillary. Interstitial hydrostatic pressure is uniform within a local region; plasma oncotic pressure is high in plasma and substantially lower in the interstitial space. Thus, there is usually net filtration of fluid out of the capillary at the arteriolar end. Toward the venular end of the capillary, where capillary hydrostatic pressure is lower, net reabsorption of fluid into the capillary usually occurs.

Excess interstitial fluid is returned to the circulation through the lymphatic system (Fig. 12.3). Movement of fluid from the



Edema is the swelling associated with an increase in interstitial fluid in a tissue or body cavity. The amount of fluid in the interstitial space depends on the rate at which interstitial fluid is produced and the rate at which it is removed. When high capillary hydrostatic pressure, low capillary oncotic pressure, or leakage of proteins from the vascular space (producing high interstitial oncotic pressure) results in excessive fluid in the interstitial space, it is ordinarily removed by the lymphatic system. Edema may occur with blockage of the lymphatic system or excessive flux of fluid from the vascular space.

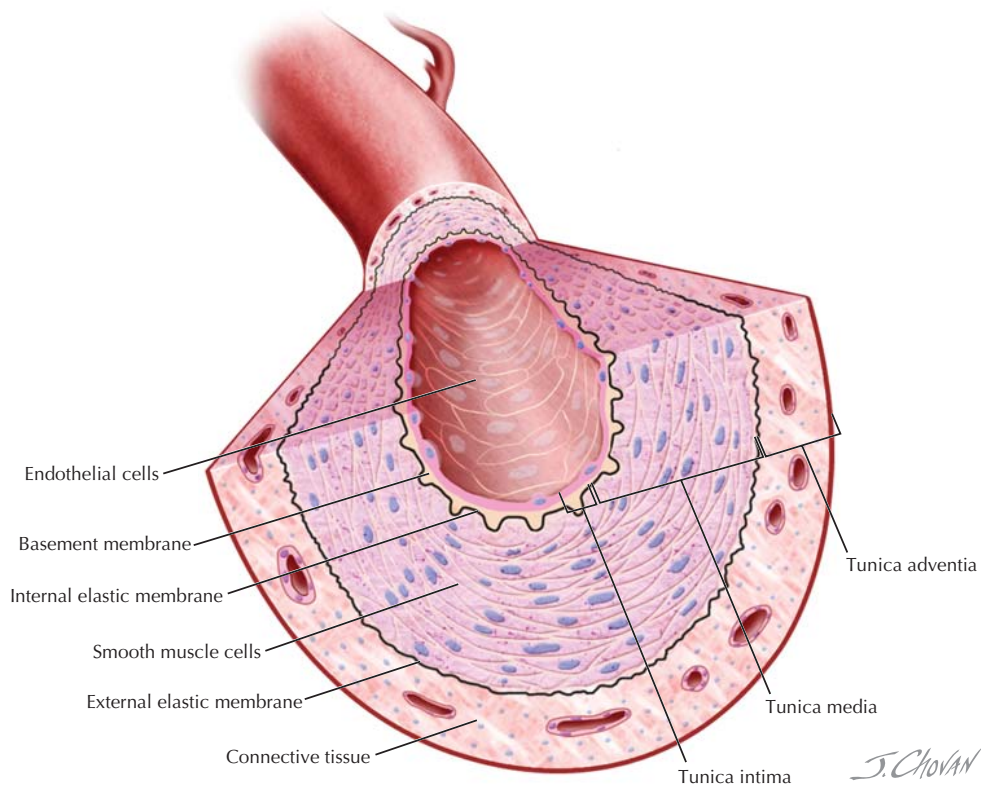


Figure 12.1 The Vascular Wall The vascular walls of arteries and veins have three tissue layers, with the relative thickness of medial and adventitial layers varying among various vessel types. The capillary wall consists of only the intimal layer of endothelial cells and basement membrane. The vessel illustrated above is a large artery, with prominent medial smooth muscle. The walls of large arteries and veins have their own vascular supply (vasa vasora). The tunica media of arteries is bounded by an internal elastic membrane (IEL) and an external elastic membrane (EEL).

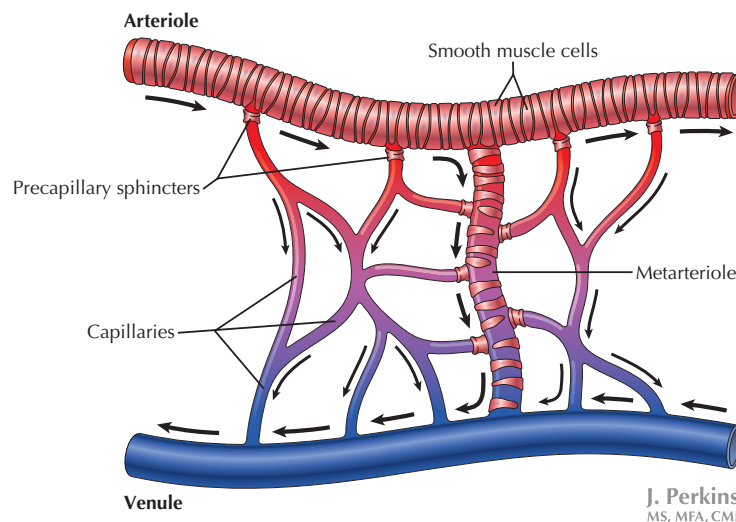


Figure 12.2 Components of the Microcirculation Arterioles, metarterioles, capillaries, and venules constitute the microcirculation. Direction of blood flow is indicated by arrows. Blood flow is regulated by constriction and dilation of smooth muscle of the arterioles, metarterioles, and precapillary sphincters. Sympathetic nerves extend to arterioles, metarterioles, and venules in the microcirculation.

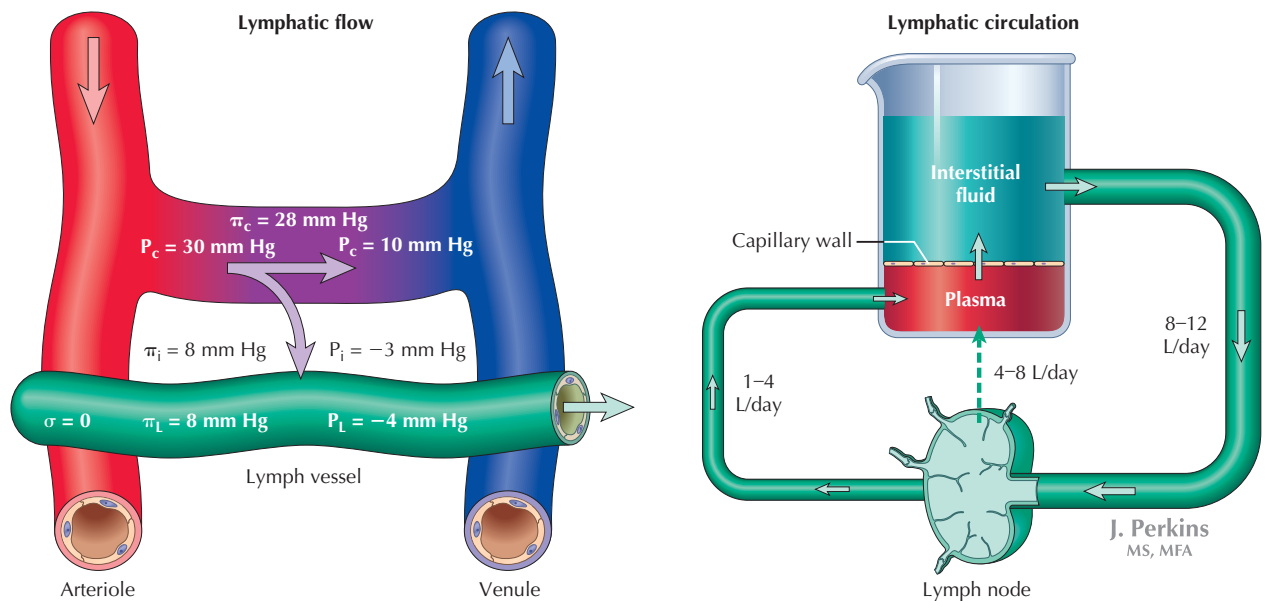


Figure 12.3 The Lymphatic Circulation Diffusion of fluid from the vascular space to the interstitial space is governed by the Starling equation (see Chapter 1). Excess fluid in the interstitial space is transported by the lymphatic system back to the central venous circulation. Movement of fluid from interstitium to the lymphatic system is driven solely by the hydrostatic pressure gradient, because there is no oncotic pressure gradient (protein flows freely through the lymphatic vessel wall). ($P_c = HP_c$, hydrostatic capillary pressure.)

interstitium into the capillaries is driven by the hydrostatic pressure gradient between the interstitial space and the lymph, because there is no significant oncotic pressure gradient (the protein reflection coefficient of lymphatic capillaries is zero), and there are one-way valves through which fluid and protein flow into lymphatic vessels. Lymph flows into larger lymphatic vessels and is returned into the central venous circulation through the thoracic duct.

REGULATION OF BLOOD FLOW

Flow through a specific tissue can be altered either by changes in perfusion pressure or changes in resistance of the arterial vessels perfusing the tissue. Obviously, when blood flow requirements of a specific tissue are altered, the most efficient mechanisms for adjusting blood flow will involve a change in regional resistance. On the other hand, when systemic metabolic needs vary, for example during exercise, regulation of flow is accomplished through changes in both systemic hemodynamics (blood pressure and cardiac output) and regional resistances, resulting in changes in both overall flow (cardiac output) and flow distribution between various tissues.

At the tissue level, both intrinsic and extrinsic factors may influence smooth muscle tone in the resistance vessels (arterioles, precapillary sphincters, and small arteries) in response to local and systemic events. Both smooth muscle and endothelial cells participate in this regulation. Basic aspects of

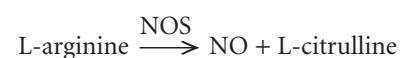
smooth muscle structure and function are discussed in Section 2.

Regulation of Vascular Tone by Endothelial Cells

The role of the endothelium in regulation of smooth muscle contraction and relaxation is a relatively recent discovery. The endothelium participates in this regulation through these processes (Fig. 12.4):

- release of vasodilators such as nitric oxide and prostacyclin
- release of the vasoconstrictors endothelin
- conversion of angiotensin I to angiotensin II

Of particular importance is the endothelium's role in vascular regulation through production of **nitric oxide (NO)**. When endothelial cells are exposed to **endothelium dependent vasodilators** such as acetylcholine, histamine, and bradykinin, the enzyme nitric oxide synthetase (NOS) is activated, resulting in the production of the short-lived but powerful vasodilator NO from the amino acid arginine:



Nitric oxide diffuses to adjacent smooth muscle cells and acts on smooth muscle guanylyl cyclase, elevating cGMP. cGMP causes reduction in free intracellular Ca^{2+} , producing

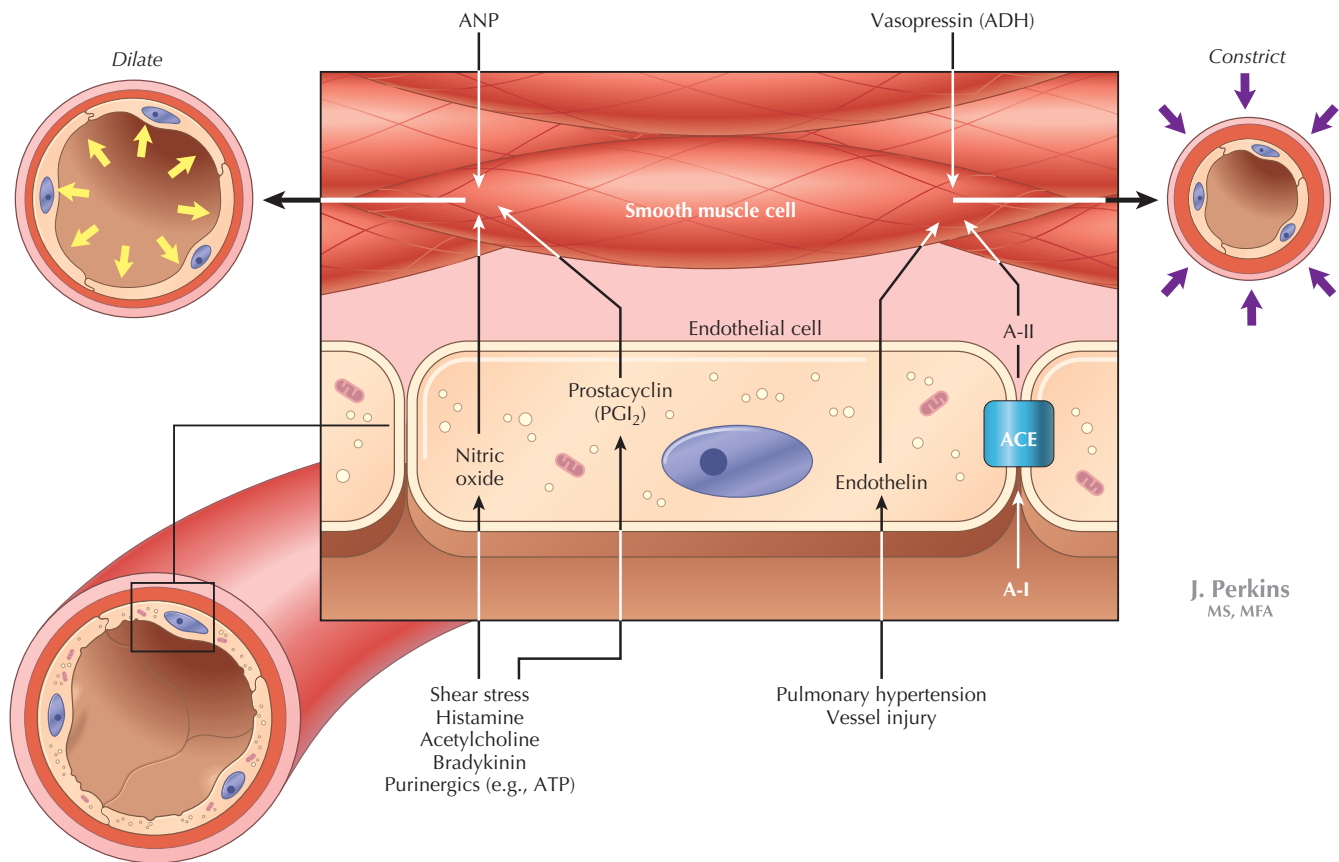


Figure 12.4 Control of Arteriolar Tone The greatest resistance to flow occurs in the small arteries and arterioles. The state of constriction or relaxation of these vessels is regulated in part by the sympathetic nervous system and the release of norepinephrine. Circulating hormones, including vasopressin (ADH) and angiotensin II (A-II), may contribute to constriction through their actions on vascular smooth muscle; atrial natriuretic peptide (ANP) is a smooth muscle dilator. The endothelium plays an important role in regulating vascular tone by its release of nitric oxide (NO) and prostacyclin (PGI₂) in response to many factors, including shear stress, acetylcholine, and bradykinin. Endothelin is a potent, endothelium-derived vasoconstrictor important in some pathophysiologic states. The endothelial cell surface also has angiotensin-converting enzyme, which forms angiotensin II by cleavage of circulating angiotensin I (A-I, an inactive precursor).

relaxation of smooth muscle and thus vasodilation. This mechanism is particularly important in vascular responses to inflammatory stimuli, as well as in parasympathetic-mediated vasodilation in the genital system and lower gastrointestinal tract (vessels in other regions are not innervated by the parasympathetic nervous system). In addition, many substances that directly act on smooth muscle cells to produce constriction (for example, norepinephrine) also release endothelial NO. This simultaneous release and action of NO results in a dampened response to vasoconstrictors in healthy vessels, as opposed to vessels in which the endothelium has been damaged. Many of the same stimuli that release NO also release prostacyclin (PGI₂) from endothelial cells, a vasodilator metabolite of arachidonic acid. Although prostacyclin is less important than NO in regulation of vascular tone, both mediators are important in preventing activation of platelets and their adherence to the vascular wall. The actions of pros-

tacyclin are opposed by **thromboxane A₂**, which is a major product of arachidonate in activated platelets. Thus, thromboxane A₂ is a potent prothrombotic substance and vasoconstrictor, important in inflammatory states and hemostasis.

The release of NO by **shear stress** is believed to participate in vascular control as follows: When blood flow to a region is increased (for example, by vasodilation in the arterial microcirculation), the higher flow causes elevated shear stress in the endothelial cells of vessels supplying the region, which respond by increasing NO production. Adjacent smooth muscle cells relax when exposed to NO, resulting in vasodilation and further augmentation of flow.

Endothelin is a powerful vasoconstrictor protein released by endothelial cells when vessels are damaged. Its role in vascular regulation in normal tissues is controversial, although it is an



The nitric oxide produced by endothelial cells is sometimes referred to as “EDRF,” or endothelium-dependent relaxing factor. This name is based on the original discovery by Robert Furchgott that vessel segments experimentally denuded of endothelium did not display smooth muscle relaxation when stimulated by acetylcholine and other “endothelium-dependent” vasodilators. Subsequent studies during the 1980s by several investigators led to the identification of EDRF as nitric oxide. The journal *Science* named nitric oxide “Molecule of the Year” in 1993, and the Nobel Prize in Medicine or Physiology went to Furchgott, Louis Ignarro, and Ferid Murad for their work in this area.

important pathophysiologic mediator, for example in pulmonary hypertension and preeclampsia. **Angiotensin-converting enzyme (ACE)** is an enzyme found on the surface of endothelial cells. This enzyme cleaves angiotensin I to **angiotensin II**, a potent vasoconstrictor. Like the posterior pituitary hormone **vasopressin**, angiotensin II is important in long-term regulation of blood pressure through its effects on renal sodium and water retention (see Section 5), and plays a role in acute responses to hypotensive crises such as hemorrhage. However, neither angiotensin II nor vasopressin is believed to be important in short-term regulation of vascular function.



Until the second half of the 20th century, the endothelium was generally believed to function mainly as a lining or barrier between blood and tissues. In the past few decades, appreciation of its role in many physiological processes has blossomed. The endothelium is important in regulating vascular tone, angiogenesis (formation of blood vessels), and the hemostatic process (healthy endothelium is “antithrombotic” due to its formation of nitric oxide and prostacyclin). In addition, the endothelium has a role in metabolism. An example is conversion of angiotensin I to angiotensin II by its angiotensin-converting enzyme. The endothelium has an important role in disease processes, for example in atherosclerosis, in which diminished endothelial nitric oxide production and alterations in other endothelial functions are among the earliest changes observed.

Local Control of Blood Flow

Local regulation of blood flow occurs through mechanisms that involve responses to local metabolic products and transmural pressure (pressure gradient across the vascular wall) (Fig. 12.5A). If blood flow to a region is occluded temporarily, when flow is reestablished, **reactive hyperemia** occurs. In other words, blood flow is elevated above the original level. Reactive hyperemia occurs as a result of **local metabolites** that accumulate during the occlusion. These metabolites, which include CO_2 , H^+ , K^+ , lactic acid, and adenosine, act directly on

smooth muscle of arterioles and precapillary sphincters in the region to produce vasodilation. Thus, flow is elevated until the tissue levels of O_2 are restored and accumulated metabolites are removed. In contrast to reactive hyperemia, **active hyperemia** refers to the increase in flow that occurs in tissues when metabolism is elevated (Fig. 12.5B). An example of this is the greatly increased flow to skeletal muscles during exercise. Ongoing production of local metabolites in the working skeletal muscles causes vasodilation, thus enhancing flow to the muscles.

Autoregulation of local blood flow can occur without changes in local metabolism (Fig. 12.5C). In many tissues and organs, if blood flow is artificially increased by raising the perfusion pressure to the vascular bed, flow will be immediately elevated as expected; however, it soon returns toward the basal rate. According to the **myogenic hypothesis**, smooth muscle cells constrict in response to elevated transmural pressure, in other words, in response to stretch. This is a mechanism whereby flow to a tissue is kept fairly constant, despite changes in pressure, when metabolic needs of the tissue are not changing. Myogenic regulation does not involve the vascular endothelium, but is a direct response of smooth muscle cells.

Extrinsic Regulation of Peripheral Blood Flow

Extrinsic regulation of peripheral blood flow involves vasoconstriction or vasodilation in response to neural mechanisms and circulating vasoactive substances. Smooth muscle cells have several types of adrenergic receptors:

- **α -Receptors**, which mediate the constrictor responses to catecholamines. Receptors of the subtype α_1 are the predominant subtype in vascular tissue; α_2 -receptors also produce vasoconstriction by inhibiting reuptake of norepinephrine. α_1 -Adrenergic vasoconstriction is mediated by the second messenger inositol trisphosphate, and α_2 vasoconstriction is mediated by reduced cAMP levels.
- **β_2 -Receptors**, which mediate dilator responses to catecholamines. β_2 -Adrenergic vasodilation is mediated by the second messenger cAMP.

When the sympathetic nerves are stimulated, the responses of vessels will be dependent on the type of adrenergic receptors activated (a function of receptor density and agonist concentration), with the predominant systemic response being vasoconstriction. The major arterial response to sympathetic nervous system activation is α -receptor-mediated vasoconstriction, for example in baroreceptor-mediated responses (see Fig. 11.2). Note that generalized arterial constriction produces elevation of arterial blood pressure, because arterioles and small arteries are the major site of action, and constriction of these vessels produces increased peripheral resistance. Sympathetic activation results in widespread constriction of veins, elevating venous pressure and enhancing preload on the heart. β_2 -receptors are present in some arterial

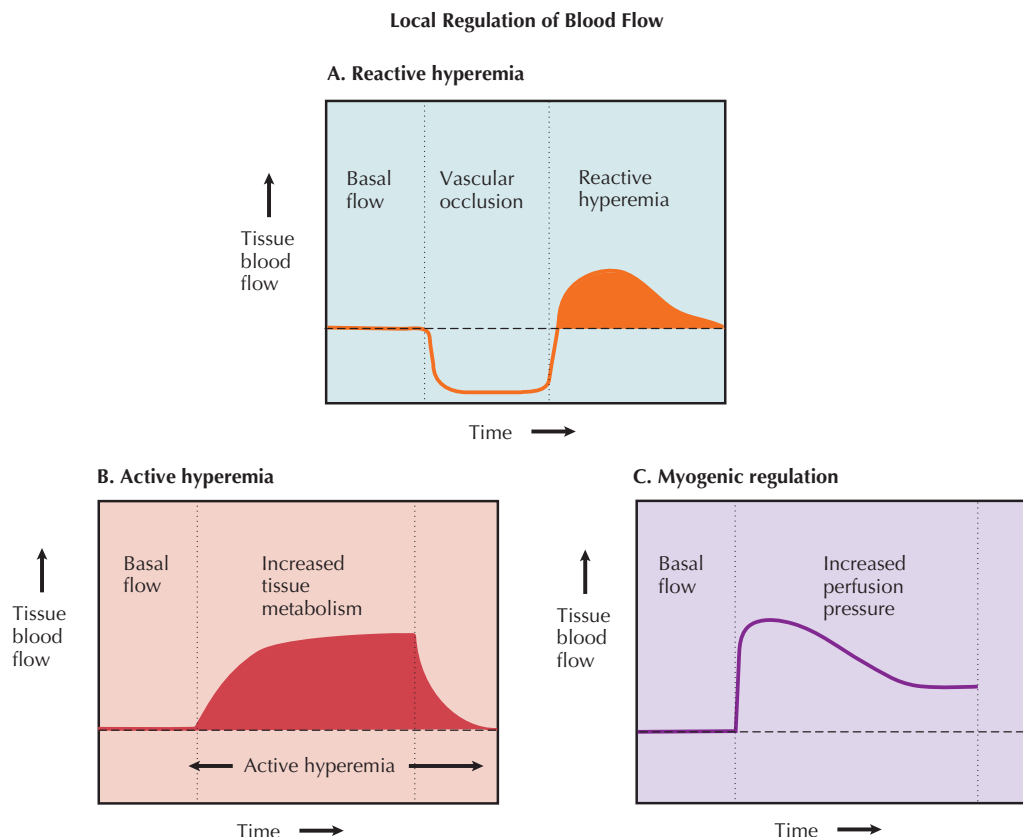


Figure 12.5 Local Regulation of Blood Flow Regional blood flow is controlled by local as well as neural and humoral factors. **A**, Reactive hyperemia occurs when blood flow is reestablished after occlusion. Buildup of vasodilatory metabolites such as CO_2 , H^+ , K^+ , lactic acid, and adenosine results in relaxation of arterioles and precapillary sphincters directly exposed to these substances. Hence, a period of increased blood flow (hyperemia) occurs upon reperfusion. **B**, Active hyperemia refers to the increase in blood flow to a tissue when metabolism in the tissue is elevated; this hyperemia is a result of increased production of vasodilatory metabolites. **C**, Myogenic regulation refers to the autoregulation of blood flow that occurs when perfusion pressure is increased (with no change in metabolic activity in the tissue). Initially, when perfusion pressure is raised, flow to the tissue rises as expected, but returns toward the baseline. Smooth muscle of the arterial microcirculation constricts in response to a rise in transmural pressure, thus autoregulating blood flow.

vessels, for example, in skeletal muscle, where sympathetic outflow in anticipation of exercise produces increased blood flow to muscle.



While the sympathetic nervous system innervates vessels in most parts of the circulation, parasympathetic innervation of blood vessels is absent in most tissues. Exceptions are the genital organs, salivary glands, and lower gastrointestinal tract. In circulations innervated by the parasympathetic nervous system, release of acetylcholine stimulates endothelial cells to release nitric oxide, producing nitric oxide-dependent vasodilation of underlying smooth muscle. Because acetylcholine is short-lived in the circulation, it affects only the vessels that are directly innervated by parasympathetic nerves.

In addition to release of catecholamines at sympathetic nerve endings in the vasculature, epinephrine may be released into the bloodstream by the adrenal medulla during sympathetic nervous system activation. In this case, epinephrine acts as a circulating hormone; other circulating hormones that affect vascular tone include angiotensin II and vasopressin.

Chemoreceptors also participate in the extrinsic regulation of vascular tone. The aortic and carotid bodies are densely vascularized groups of specialized cells found in the proximity of the arterial baroreceptors. Although they are more important in regulation of respiration (see Section 4), they respond to reduced arterial PO_2 , and to a lesser extent to elevated PCO_2 and H^+ concentration, by initiating a neural reflex that produces arterial and venous constriction.

REGULATION OF ARTERIAL BLOOD PRESSURE

From the foregoing, it should be clear that local, intrinsic mechanisms for vascular regulation are mainly aimed at regulation of regional blood flow, whereas neural and humoral mechanisms are often aimed at regulation of arterial blood pressure. The maintenance of mean arterial pressure (MAP) near its normal level is necessary for adequate perfusion of tissues throughout the systemic circulation. Rearranging the flow equation ($Q = \Delta P/R$), the relationship $\Delta P = R \times Q$ results, where ΔP is the pressure gradient, R is resistance, and Q is flow. The pressure gradient is the difference between arterial and central venous pressure. For the overall systemic circulation, Q is equal to cardiac output and R is total peripheral resistance (TPR), yielding

$$\Delta P = CO \times TPR$$

Because venous pressure is very low, this equation can be essentially reduced to:

$$MAP = CO \times TPR$$

Thus, regulation of arterial blood pressure involves regulation of cardiac output and systemic resistance, through the various mechanisms already described. Blood pressure is monitored at several points in the system (Fig. 12.6):

- aortic arch and carotid sinus baroreceptors
- renal juxtaglomerular apparatus
- low-pressure (cardiopulmonary) baroreceptors

Short-Term Regulation of Blood Pressure by Arterial Baroreceptors

The **high-pressure arterial baroreceptors** of the aortic arch and carotid sinus and the associated baroreceptor reflexes are most important for moment-to-moment regulation of arterial pressure. During normal, quiet daily activities, an inverse relationship is usually observed between changes in blood pressure and heart rate, reflecting this role of the baroreceptor reflex in maintaining pressure: when pressure falls, heart rate rises; when pressure rises, heart rate falls. These fluctuations in heart rate reflect changes in sympathetic and parasympathetic outflow from the medullary cardiovascular centers, in response to the degree of baroreceptor stretch.

Efferent sympathetic nerve activity is inversely related to MAP between the range of 60 to 160 mm Hg in a normal individual (see Fig. 11.2). At pressures below 60 mm Hg, sympathetic nerve activity is maximal. In addition to MAP, baroreceptors are also sensitive to pulse pressure. If pulse pressure is dampened while MAP is held constant, afferent nerve impulses from the baroreceptors to the cardiovascular center will be less frequent, and sympathetic efferent activity and thus arterial blood pressure will be elevated.

The arterioles of the **juxtaglomerular apparatus** of the kidneys also contain high-pressure baroreceptors; in this case, stretch



“Resetting” of arterial baroreceptor sensitivity occurs when blood pressure is chronically elevated. In a healthy person with normal blood pressure, baroreceptors function to return arterial pressure to the normal level when pressure falls or rises. With chronic hypertension, a new set-point is established, whereby baroreceptor activity is aimed at maintaining the higher resting blood pressure. Although this change appears, on the surface, to be maladaptive, it allows adequate short-term regulation of blood pressure despite higher basal blood pressure.

results in release of the enzyme renin by the kidney. Renin enzymatically cleaves the plasma protein angiotensinogen (a liver product) to form angiotensin I, which is converted to angiotensin II by endothelial angiotensin-converting enzyme. This mechanism is important in short-term regulation of blood pressure only during pathophysiologic states such as hemorrhage.

Role of Low-Pressure Baroreceptors and Atrial Stretch

Low-pressure baroreceptors are found in low-pressure sites of the circulation, specifically the atria and large vessels of the pulmonary circulation, and respond to changes in blood volume. The Bainbridge reflex was discussed earlier; increased atrial stretch initiates a reflexive increase in heart rate. Reduction of blood volume, for example during hemorrhage, is detected by low-pressure baroreceptors in the left atrium, as well as arterial baroreceptors. Neural afferent signals through the vagus nerve to the hypothalamus result in vasopressin release by the posterior pituitary. Like angiotensin II, vasopressin participates in short-term responses to hemorrhage, but not in acute regulation of blood pressure under normal circumstances.

Increased blood volume and stretch of atria stimulates atrial myocytes to secrete stored **atrial natriuretic peptide (ANP)**. ANP dilates some vessels (although it has little role in acute blood pressure regulation) and has important effects on sodium and water balance (see Section 5) and long-term regulation of blood pressure (see following section).

Long-Term Regulation of Blood Pressure

In contrast to moment-to-moment regulation of blood pressure, which relies heavily on baroreceptor reflexes and adjustments to cardiac and vascular function, **long-term regulation of blood pressure** is accomplished mainly by mechanisms that control blood volume through neural and humoral pathways (Fig. 12.7). These mechanisms are covered in further detail in Sections 5 and 7. Briefly, when blood volume is depleted and blood pressure is consequently reduced, direct effects of reduced renal perfusion, as well as increased sympathetic nerve activity, stimulate renin production by the kidneys

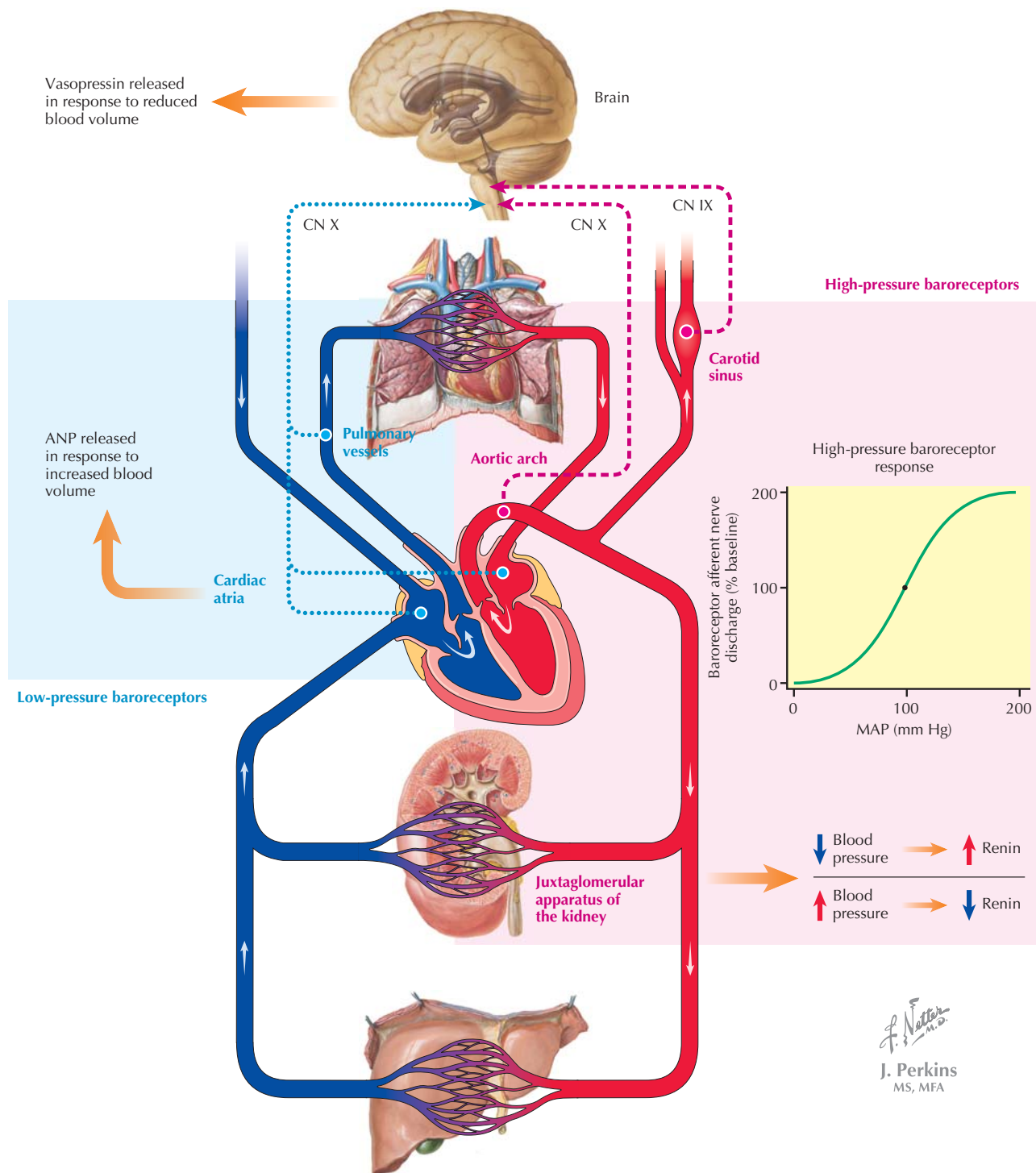


Figure 12.6 Monitoring of Blood Pressure To maintain adequate blood flow to tissues, the body has a complicated system for monitoring and regulating blood pressure. High-pressure baroreceptors in the aortic arch and carotid sinus are extremely important in acute regulation of blood pressure, through their effects on the autonomic nervous system. Afferent arterioles in the renal juxtaglomerular apparatus also contain high-pressure baroreceptors; these are involved in regulation of renin release, and consequently, regulation of sodium and water balance, important in long-term regulation of blood pressure. Low-pressure baroreceptors in the heart and pulmonary circulation respond to changes in blood volume and modulate sympathetic activity and vasopressin release. The cardiac atria also release atrial natriuretic peptide (ANP) in response to elevated blood volume.

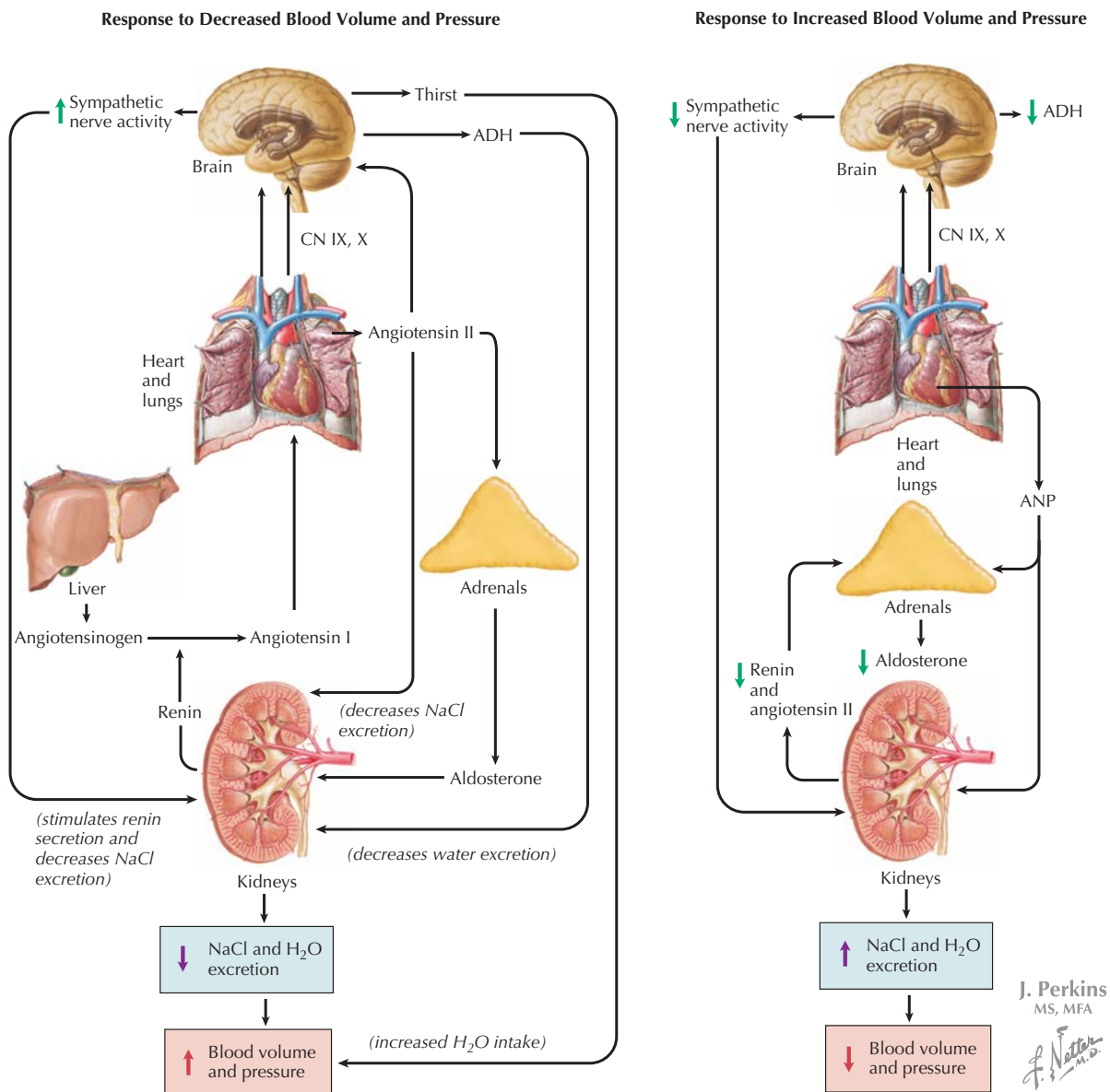


Figure 12.7 Long-Term Response to Changes in Blood Volume and Pressure In addition to evoking mechanisms for acute adjustment of blood pressure, changes in blood volume and pressure will also activate renal mechanisms for adjusting blood volume. Reduced blood volume (and therefore arterial pressure) will stimulate the renin-angiotensin-aldosterone system, with the end result of sodium and water retention. Reduced blood pressure will also activate the sympathetic nervous system, which will stimulate renin secretion as well as have direct effects on the kidneys. On the other hand, increased volume will stimulate atrial natriuretic peptide (ANP) release by the heart. ANP has direct renal effects (natriuresis and diuresis) and also inhibits aldosterone release by the adrenal medulla.

(sympathetic activation occurs in part due to baroreceptor responses). Renin action produces elevation of angiotensin I, which is subsequently cleaved to angiotensin II by endothelial cell angiotensin-converting enzyme (much of this enzymatic cleavage occurs in the pulmonary circulation). **Angiotensin II** has direct effects on sodium retention by the kidney and has the important effect of stimulating aldosterone release by the

zona glomerulosa of the adrenal cortex. **Aldosterone** also promotes renal sodium (and hence fluid) retention. Low blood volume, through reflexes discussed earlier, stimulates posterior pituitary antidiuretic hormone (ADH; also known as vasopressin). ADH release, as well as thirst, is also stimulated by elevated plasma osmolality associated with volume depletion. ADH promotes water retention by the kidney.